The Journal of Positive Psychology
Dedicated to furthering research and promoting good practice

ISSN: 1743-9760 (Print) 1743-9779 (Online) Journal homepage: http://www.tandfonline.com/loi/rpos20

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To cite this article: Vincent Ng, Louis Tay & Lauren Kuykendall (2017): The development and validation of a measure of character: The CIVIC, The Journal of Positive Psychology

To link to this article: http://dx.doi.org/10.1080/17439760.2017.1291850

Published online: 28 Feb 2017.

THE JOURNAL OF POSITIVE PSYCHOLOGY, 2017
http://dx.doi.org/10.1080/17439760.2017.1291850

The development and validation of a measure of character: The CIVIC

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ABSTRACT
The present research study sought to develop and validate a character scale – the Comprehensive Inventory of Virtuous Instantiations of Character using a total sample size of 3679 across five studies. In Study 1, character trait items were generated using an integrative classification system. In Study 2, character trait scales were further refined and their factor structure examined, revealing eight higher-order character dimensions or character cores: appreciation, intellectual engagement, fortitude, interpersonal consideration, sincerity, temperance, transcendence, and empathy. Study 3 established convergent validity of character traits with extant measures and discriminability from personality facets, social desirability, and moral cognitive development. Study 4 revealed that character cores were more strongly related to evaluative constructs than personality dimensions. Study 5 demonstrated that character cores predicted performance and psychological well-being outcomes above and beyond personality. The implications of our findings for the assessment and taxonomy of character are discussed.

ARTICLE HISTORY
Received 9 February 2016
Accepted 13 January 2017

KEYWORDS
Virtue; character; strengths; personality; morality; scale development; measurement; exploratory structural equation modeling; subjective well-being; eudaimonia

Since its inception, positive psychology has sought to study the constituents of and conditions that engender optimal human functioning (Gable & Haidt, 2005; Wong, 2011). In part this has included the cultivation of certain positive traits or strengths of character (Peterson & Seligman, 2004; Seligman & Csikszentmihalyi, 2000). The growing attention on character strengths brings a critical question into focus: what is the structure of good character? Addressing this question is important for both scientific and practical purposes. From a scientific perspective, understanding the structure of character is foundational to both construct and predictive validity. The scientific advancement of substantive theories in this burgeoning area of research depends on elucidating the key dimensions of character and how they differentially predict outcomes (Park, Peterson, & Seligman, 2004; Shimai, Otake, Park, Peterson, & Seligman, 2006). From a practical standpoint, a taxonomy of character traits would serve as the positive analog to the Diagnostic and Statistical Manual (DSM-IV) (American Psychiatric Association, 2000) (see Dahlsgaard, Peterson, & Seligman, 2005) by explicating a coherent theory about what is right in people as a starting point for conceptualizing mental disorders as the opposite, absence, or excess of these strengths of character (Seligman, 2014). From a practical standpoint, this unDSM could serve as the basis for the diagnosis and evaluation of interventions intended to develop good character (Peterson, 2006).This has deep societal importance for the populace at large who do not suffer from clinical mental health issues.

At present, the structure of character strengths stems from the theoretical foundation of the VIA Classification of Strengths and Virtues (CSV; Dahlsgaard et al., 2005; Peterson & Seligman, 2004). The CSV is a theoretically derived classification scheme of six broad, abstract categories of morally-valued core characteristics (i.e. virtues) that emerged consistently from in-depth textual reviews of the world's most influential religions and philosophies: wisdom, justice, temperance, courage, humanity, and transcendence (Dahlsgaard et al., 2005). Peterson and Seligman (2004) intentionally structured the CSV hierarchically because the abstract virtues were deemed too general to measure directly. That is, they and a core group of scholars proposed potential discrete traits (i.e. character strengths) that would represent distinct manifestations of each of the six core virtues from a review of psychological and other literature using 10 criteria (see Peterson & Seligman, 2004). Crucially, their comprehensive classification scheme aligns conceptually fairly well with preliminary factor structures revealed using preliminary empirical bottom-up approaches (Cawley, Martin, & Johnson, 2000). As shown in Table 1, the integrative classification separates the Cawley et al.'s (2000) lexical dimension of Order into Justice and Temperance as well as the lexical dimension of Resourcefulness into Courage and Wisdom. As such, using the CSV as a conceptual foundation for

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Table 1. Conceptual similarity between empirical and theoretical virtue dimensions.

Cawley et al. (2000)					
Factor 1: empathy	*Factor II: order*		*Factor III: resourceful*		*Factor IV: serenity*
Empathy	Order		Resourceful		Serene
Concern	Discipline		Purposeful		Meek
Understanding	Serious		Perseverance		Forbearance
Considerate	Decent		Persistence		Forgiveness
Friendly	Deliberate		Confidence		Peaceful
Sympathy	Scrupulous		Sagacity		Patient
Affable	Earnest		Self-esteem		
Sensitive	Self-control		Fortitude		
Charity	Self-denial		Intelligence		
Compassion	Abstinence		Zealous		
Liberal	Obedient		Independent		
Gracious	Conservative				
Courtesy	Cautious				
	Careful				
	Tidy				
	Austere				
	Clean				
Peterson and Seligman (2004); Dahlsgaard et al. (2005)					
Humanity	*Justice*	*Temperance*	*Courage*	*Wisdom*	*Transcendence*
Love	Fairness	Forgiveness	Bravery	Creativity	Gratitude
Kindness	Leadership	Prudence	Honesty	Curiosity	Hope
Social intelligence	Teamwork	Humility	Perseverance	Judgment	Appreciation of beauty
		Self-control	Zest	Perspective	Humor
				Love of learning	Spirituality

scale creation would constitute an integrative approach at present, though requiring empirical development and validation.

Indeed, this classification scheme is considered a theoretical one that is open to revision and interpretation. Peterson and Seligman (2004) noted that they have made a 'good start in measuring strengths of character, but much of the journey remains' (p. 625). Peterson and Park (2009) further state, 'Our classification of character strengths under core virtues is a conceptual scheme and not an empirical claim' (p. 31). Indeed, Peterson and Seligman (2004) urged readers to, '… not to be too concerned about the details of how we classified the 24 strengths under the six virtues. We have not measured the virtues per se; they are too abstract and general' (p. 31).

The seminal development of a measure based on the CSV, the Values in Action-Inventory of Strengths (VIA-IS; Peterson & Seligman, 2004), has enabled a preliminary understanding of the structure of virtues and character. Nonetheless, there are questions as to the measurement adequacy of the VIA-IS. Foremost, the factor structure of virtues using the VIA-IS has not shown clear convergence across samples, ranging from as few as one (Singh & Choubisa, 2009) to as many as five factors (Littman-Ovadia & Lavy, 2012; Peterson, Park, Pole, D'Andrea, & Seligman, 2008; Ruch et al., 2010; Singh & Choubisa, 2010), the five factor structure being the most prevalent (see Table 2). This is problematic because the lack of empirical consensus on the structure of good character indicates measurement issues. One major potential reason for this is that the scales are not unidimensional (Duan et al., 2012; Khumalo, Wissing, & Temane, 2008; McGrath, 2014), which can lead to a distorted factor structure (Hopwood & Donnellan, 2010). Indeed, recent research using a large data-set indicated that none of the 24 character strengths were unidimensional (Ng, Cao, Marsh, Tay, & Seligman, 2016). Another contributing element is the use of principal components analysis as the typical extraction method used in the analysis of the factor structure of the VIA-IS, which is inappropriate for modeling the constructs as reflective latent factors (Borsboom, 2006). As such, the VIA-IS measure, while groundbreaking and vital for promulgating virtues and positive psychology research, may not be adequate for addressing key questions on the empirical taxonomy of virtues.

Measurement issues aside, the VIA-IS needs further construct validation. Noftle, Schnitker, and Robins (2011) found character strengths correlated highly with personality facets and argued on the basis of these results that future research needs to empirically demonstrate character's distinctiveness from personality. Indeed, other researchers have found moderate to strong relationships between character strengths and personality measures (e.g. Macdonald, Bore, & Munro, 2008; Steger, Hicks, Kashdan, Krueger, & Bouchard, 2007) and it is thus essential to the construct validation process to show that character is not simply redundant with current taxonomies of personality. It is important to note at this juncture that the empirical question we pursue in the remainder of this article is whether character is distinct from and predicts outcomes beyond personality as measured using the dominant personality taxonomy researchers use as a

Table 2. Comparison of derived factor structures from analysis of VIA scales scores.

						Conceptual overlap with Peterson and Seligman's (2004) six virtues					
Authors (date)	*N*	Language	Sample	Extraction method	Retention rule (# of factors)	Justice	Temperance	Courage	Wisdom	Transcendence	Humanity
Brdar and Kashdan (2010)	881	Croatian	Croatian students	PAF	MAP, Int. (4)	*Interpersonal strengths*	Cautiousness		Fortitude	*Vitality*	
Khumalo et al. (2008)	256	English	African students	PCA	K1 (3)	*Integrity in a group context*			*Intrapersonal and relational strengths*		
Littman-Ovadia and Lavy (2012)	635	Hebrew	Israeli adults	PCA	K1, scree (5)	*Interpersonal strengths*		*Strengths of restraint*	Intellectual strengths	Theological strengths	Emotional strengths
Macdonald et al. (2008)[a]	123	English	Australian students	PCA	Int. (4)		*Conscientiousness*		Intellect	Niceness	Positivity
McGrath (2014)	458,988	English	US adults (online)	PAF	PA, MAP (5)	Interpersonal	Restraint	*Emotional*	Intellectual	Theological	
Peterson and Park (2004)	~75,000	English	US, Canadian, UK, and Australian adults	EFA	(5)	Interpersonal strengths	Conative strengths		Cognitive strengths	Transcendence strengths	*Emotional strengths*
Peterson et al. (2008)	1739	English	Mostly US adults	PCA	K1 (5)		Temperance	Fortitude	Cognitive	Transcendence	Interpersonal
Peterson and Seligman (2004)		English		EFA	(5)	Interpersonal strengths	Restraint strengths		Intellectual strengths	Theological strengths	*Emotional strengths*
Ruch et al. (2010)	1674	German	German adults	PCA	K1 (5)	*Interpersonal strengths*	*Strengths of restraint*		Intellectual strengths	Theological strengths	*Emotional strengths*
Ruch, Martín-ez-Martí, Proyer, and Harzer (2014)[b]	211	German	German adults	PCA	Scree (5)		Strengths of restraint	Emotional strengths	Intellectual strengths	Theological strengths	*Interpersonal strengths*
Singh and Choubisa (2009)	186	Hindi	Indian students	PCA	K1, scree (1)	Unidimensional factor					
Singh and Choubisa (2010)	123	English	Indian students	PCA	K1, scree (5)	Civic strengths		*Self-assurance strengths*	Intellectual strengths	Theological strengths	Interpersonal strengths
Shryack, Steger, Krueger, and Kallie (2010)	332	English	US twin adults	PCA	PA, scree, BA (3)	Sociability	Conscientiousness		Agency/self-assuredness		
					Component identification (4)	Sociability	Conscientiousness		Agency/self-assuredness	Serenity	

Notes: Factors in italics conceptually overlapped equally with two or more virtues from Peterson and Seligman's (2004) six virtue taxonomy. Samples designated as student samples included only students; adult samples potentially included students. Study with [a]indicates study used the International Personality Item Pool version of the VIA-IS; study with [b]indicates study used a short form (24 item) version of the VIA-IS. PCA = principal components analysis, PAF = principal axis factoring, EFA = exploratory factor analysis, MAP = minimum average partial procedure, Int. = interpretability of factor, K1 = Kaiser's criterion of retaining factors with eigenvalues greater than one, scree = scree plot inspection, PA = parallel analysis, BA = Goldberg's (2006) 'bass-ackwards' technique, component identification requires that factors have at least three indicators loading strongly on it for retention.

summary of normal personality in the Big Five (Goldberg, 1990; Saucier & Srivastava, 2015), not whether character resides outside the entire domain of all non-cognitive-ability individual differences, of which character is clearly a part. Relatedly, Noftle et al. (2011) further stated that little work has been done to establish convergent validity of character strengths with extant individual difference measures and situate them in a nomological net (Cronbach & Meehl, 1955). Indeed, most of the research on the VIA-IS has focused on character strengths' predictive validity, particularly with various conceptions of well-being (Niemiec, 2013) with few notable exceptions examining their association with related constructs (e.g. Haslam, Bain, & Neal, 2004). And with regard to predictive validity, Noftle et al. (2011) questioned the utility of character strengths in incrementally predicting salient outcomes (e.g. well-being) beyond personality traits after their results indicated inconsistent support. These oversights need redressing to establish any measure of character and the virtues that describe it as construct valid.

In view of the conceptual strengths of the CSV, but measurement inadequacies of the current VIA-IS, the aim of this article is to use the theoretical foundation of the CSV to develop and validate an improved and integrative measure of character strengths for research. A strong measurement foundation would not only provide a clear empirical structure of character, but would also be fundamental for scientific progress on this topic (Torgerson, 1958). Our paper contributes to the extant literature in three major ways: (a) by providing an integrative measure of character traits that has good psychometric properties (e.g. unidimensional indicators); (b) by using the most appropriate statistical methods on the data collected to reveal a clearer, more consistent factor structure; and (c) establishing character's nomological net by examining how discriminable character is from personality and contrasting their differential relations with related constructs.

Defining character, virtues, and character traits: Conceptual considerations

From a psychological perspective, defining character requires an understanding of what personality is. Funder (2001) defined personality as '… individuals' characteristic patterns of thought, emotion, and behavior together with the psychological mechanisms – hidden or not – behind those patterns' (p. 198). Personality essentially includes any and all individual differences and their related psychological mechanisms. Allport (1937) famously stated in his theory of personality that, 'Character is personality evaluated, and personality is character devaluated' (p. 52). Seeking to provide greater clarity on the distinction between character and personality, Fleeson, Furr, Jayawickreme, Meindl, and Helzer (2014) chronicled the various definitions of character used within psychology and suggested that character encompasses a subset of personality characteristics that are relevant for their evaluative – particularly moral – significance. Thus, character can be understood as a superordinate construct that is described by certain evaluative traits which are noteworthy for their moral relevance and have classically defined as moral excellences and called virtues (Irwin, 1999; McCullough & Snyder, 2000).

Virtues have been defined in two parallel ways. First, McCullough and Snyder (2000) defined and contrasted the virtues as 'any psychological process that enables a person to think and act so as to benefit him- or herself and society' (p. 1) against capitalized 'Virtue', which are global character evaluations. Unlike 'Virtue', virtues are interpreted as underlying *psychological mechanisms* that promote good. Second, Peterson and Seligman (2004) proposed various levels of abstraction of virtues. In their definition, virtues are the core characteristics that have been pan-culturally prized by moral philosophers and religions, whereas *character strengths* are the more concrete 'psychological ingredients' of virtues that manifest in the form of distinguishable traits (p. 13). Character strengths are essentially virtues as defined by McCullough and Snyder (2000). We concur that virtues are broad ideal dispositions of character that have been cross-culturally valued in their own right, contribute to individual and societal thriving, and are exhibited in more discrete character traits.

The question that follows from these definitions is what are the characteristics that have been proposed to be virtuous? Dahlsgaard et al.'s (2005) cross-cultural analysis of religious and philosophical texts yielded six core virtues that appeared to be ubiquitous: courage, justice, temperance, wisdom, humanity, and transcendence. Potential discrete manifestations (i.e. character strengths) of each of these virtues were then sought out for nomination through an exhaustive review of relevant scholarly (i.e. psychiatry, youth development, philosophy, and psychology) and applied (i.e. character education programs and strengths-based social work) fields of inquiry, inventories of virtues or strengths by historical figures (e.g. Benjamin Franklin, Sir John Templeton), and strengths indirectly suggested by messages and slogans in popular culture (e.g. Hallmark cards, bumper stickers, and residence halls of Hogwarts; Peterson & Seligman, 2004). As Peterson and Seligman (2004) stated, 'Our intent was to leave no stone unturned in identifying candidate strengths for the classification' (p. 15). Potential character strengths were evaluated on the basis of 10 criteria for inclusion (e.g. intrinsically valued despite tendency to lead to good outcomes, contributes to fulfillment of the good life for self and others, society has institutions and related rituals to encourage their cultivation, displaying it tends to elevate rather than diminish witnesses, etc.) resulting in 24 strengths arrayed under the six virtues. Peterson and Seligman (2004) noted that although many of the 24 traits had been already been researched to some extent in the psychological literature, they tended to be researched in isolation from each other. The CSV taxonomy represents a theoretically grounded account and integrative account of character traits that undergird the broader six virtues. Thus, we recognize by building on that framework we are measuring *virtuous* instantiations of character. We define the individual differences assessed as character traits and refer to the factors that empirically emerge from their measurement as character cores to distinguish the latter conceptually from the six core virtues found by Dahlsgaard et al. (2005): a taxonomy of good character is what we seek.

Overview of studies

To develop a psychometrically sound and integrative measure of character, we first generated items to tap virtuous character traits based on the integrative classification of strengths (Peterson & Seligman, 2004), established unidimensionality of the character trait scales, and explored the factor structure that emerged from their analysis (Studies 1–2). We then confirmed the factor structure using the statistical techniques most well-suited for this type of data and investigated character traits' and cores' discriminability from personality facets and domains, respectively, by examining their differential relations with relevant constructs (Studies 3–4). Finally, we established character cores' utility by demonstrating their incremental validity in predicting performance and well-being beyond personality (Study 5).

Study 1: Preliminary item generation, selection, and construct unidimensionality

The CSV proposed by Peterson and Seligman (2004) represents a rich and integrative account of the broad human characteristics that are widely valued. We used the 24 character strengths delineated and defined within the classification scheme as a basis from which to iteratively generate and refine items (Hinkin, 1998). Emphasis was placed on ensuring items were direct, clear, consistent in presentation, and broad (i.e. cutting across domains and behaviors) as was appropriate for the conceptualization of character traits as dispositional tendencies (Clark & Watson, 1995).

Method

Five research assistants trained on item writing engaged in item generation on a weekly basis. Each item writer created approximately 10 items per week for each of the 24 character strengths using conceptualizations found in *Character Strengths and Character cores* (Peterson & Seligman, 2004) as a starting point. Items were written to emphasize the affective, cognitive, and behavioral orientation of the character trait (e.g. appreciation of beauty items were written in more affective and cognitive terms than behavioral). Instructions were given to item writers to avoid generating items that would likely lead to decreased score variability in respondents. The item writing process yielded an initial pool of 1200 potential items. A Ph.D. student well versed in scale creation and the virtues literature selected 360 items on the basis of item reading level, clarity, and operationalization of virtues. Each character trait scale was comprised of 15 items (five items per low, medium, and high levels of anticipated base rate endorsement). All items were measured on a 5-point Likert scale ranging from 1 ('Very much unlike me') to 5 ('Very much like me'). These were included for a pilot study with undergraduates to assess unidimensionality of each of the scales.

Participants and procedure

Data was initially collected from 546 students from a Midwestern university. Outliers in survey completion time, those who 'straight lined' (i.e. gave the same response for all items), and those who had more than 20% responses missing (total or for any scale) were excluded from further analysis. There were 500 students in the final sample (203 females, $M_{age} = 19.63$, $SD_{age} = 1.87$). Participants completed an online survey that presented character trait items followed by demographic variable questions. Items were presented to participants in randomized order to control for the potential of item-order effects (Knowles, 1988; Knowles & Byers, 1996; but cf., Schell & Oswald, 2013). We took this precaution to ensure the factor structure that emerges from analysis of the randomly-ordered character trait item data is more likely due to the actual structure of latent constructs rather than a side effect of the way in which items were presented.

Results

The unidimensionality of scales developed for each character trait were assessed using confirmatory factor analysis (CFA) with Mplus 7.0 by fitting a one-factor solution to each (Muthén & Muthén, 2010; Simms, 2008). If the measurement model failed to meet conventional minimally acceptable fit criteria for unidimensionality (e.g. RMSEA < 0.08, TLI > 0.90, and CFI > 0.95; Marsh, Hau, & Wen, 2004) items were deleted until model fit was satisfactory. The general process of item trimming involved removing items than had factor loadings smaller than 0.60. If model fit

Table 3. Initial EFA of character traits yielding multiple factors.

Character trait	Factor solutions	EFA fit indices				Multidimensional character trait	Factors emerging from selected factor solution	Prototypical item
		CFI	TLI	RMSEA	SRMR			
Prudence	1	0.912	0.897	0.079	0.046	Prudence	Carefulness	I usually think through what I say before I say it
	2	**0.953**	**0.936**	**0.062**	**0.031**		Forethought	I always consider the long term consequences of my actions
Self-regulation	1	0.896	0.879	0.08	0.051	Self-regulation	Self-control	I can resist temptations
	2	**0.966**	**0.954**	**0.05**	**0.028**		Willpower	I can make myself persist on a difficult task
Spirituality	1	0.855	0.831	0.153	0.076	Spirituality	Spirituality	I use my spirituality as a source of support and guidance everyday
	2	**0.945**	**0.923**	**0.103**	**0.024**		Meaning/purpose	I believe my life fits into a larger purpose
Judgment	1	0.722	0.675	0.125	0.086	Judgment	Perspective-taking	I am willing to listen to different viewpoints other than my own
	2	**0.896**	**0.952**	**0.083**	**0.045**		Openness to evidence	I am always willing to change my mind in light of new evidence
Honesty	1	0.825	0.796	0.119	0.067	Honesty	Authenticity	I never portray myself falsely
	2	**0.921**	**0.891**	**0.087**	**0.043**		Trustworthiness	I always keep my promises
Social intelligence	1	0.716	0.668	0.135	0.091	Social intelligence	Social perceptiveness	I have a real knack for knowing what people are really after
	2	0.928	0.901	0.074	0.044		Propriety	I know how to act in most social situations
	3	**0.971**	**0.952**	**0.051**	**0.025**		Emotional self-awareness	I have a good understanding of my own feelings

Note: Bolded factor solutions and fit indices indicate the factor solution retained for that character trait.

remained unsatisfactory, item factor loadings below 0.70 were deleted.

Of the 24 character trait scales, 18 eventually achieved satisfactory fit indices for unidimensionality after item trimming. Six traits did not meet adequate fit criteria and could not be item trimmed adequately to satisfy the criteria. For these scales, exploratory factor analysis (EFA) was used to investigate our hypothesis that some of the 24 character traits are non-unidimensional, i.e. tapping into distinct, but related constructs we intended to ultimately refine and retain.[1] Table 3 displays the factors chosen to be retained from the EFAs of these six character traits. Most of these non-unidimensional scales appeared to assess two distinguishable character traits except social intelligence, which appeared to be tapping into three.

Discussion

As originally hypothesized, there were several character traits that appeared to tap into associated but distinguishable constructs. This corroborates the extant individual differences literature indicating that constructs similar to spirituality (Piedmont, 1999), self-control (Tangney, Baumeister, & Boone, 2004), and social intelligence (Joseph & Newman, 2010; Peterson & Seligman, 2004) in our measure are multidimensional and have similar facets to their measurement.

Study 2: Item refinement and factor structure

Having uncovered the multidimensionality of several character traits, we sought to fulfill three aims. First, we wanted to confirm their unidimensionality after item selection. Second, it was important to ensure adequate construct coverage via another round of item generation and then confirm unidimensionality in a new sample. Third, we aimed to explore the factor structure that would emerge from the full set of unidimensional character traits. This would accomplish the theoretical and practical purposes of advancing an empirically supported taxonomy of character as well as refining a reliable and valid measure for their assessment.

The six multidimensional character traits revealed in the prior study had their respective items distributed into separate scales to reflect the unidimensional scales suggested by the EFA. Each scale was refined via CFA through an iterative process by dropping lowest loading items until adequate model fit for unidimensionality was reached. This item-trimming process resulted in some character traits scales (e.g. trustworthiness) with inadequate construct coverage. Two graduate students well-versed in virtues engaged in another round of item generation targeted towards these traits. The full set of items, including the newly generated ones, was given to an expert in scale creation and psychometrics. Items were revised for reading comprehension level, clarity, to ensure greater variability in responses (e.g. replaced 'always' with 'sometimes' or 'usually'), logical consistency, and item format (i.e. each item changed to begin with 'I …') to prevent the possibility of item format from affecting responses. Items were then further generated to flesh out the constructs. This process yielded a total of 306 items capturing the 31 character traits that was then administered to the current study's sample.

Participants and procedure

Participants were recruited through Amazon's MTurk platform and compensated \$1.00 to complete the survey. Only people who lived in the US, were 18 years of age or older, and had earned at least a 95% approval rating were able to participate. After removing duplicate responses from the same individual and responses that took too little or too much time, had incorrect answers for attention check questions, or were missing more than 20% of either total responses or scale responses for any given scale, we arrived at a final sample of 1965 (1234 females; M_{age} = 35.53, SD_{age} = 12.53).

Measure

A total of 306 items measuring 31 distinct character traits (9–10 items per character trait) from the Study 1 were used to explore the higher order factor structure of character. Participants were asked to rate these items on a 4-point Likert scale with response options ranging from 1 ('Very much unlike me') to 4 ('Very much like me'). The decision was made to change the response options by removing the neutral or middle option to ensure that scores were more widely distributed across the possible options.

Results and discussion

Reliability and unidimensionality

Unidimensionality was assessed using CFA for each character trait using Mplus 7.0. It was important to assess each character trait for unidimensionality given the item revisions and additions that occurred prior to data collection for this study. Each of the character traits originally found to be unidimensional in Study 1 were re-assessed using CFA; lowest loadings were eliminated repeatedly until each character trait scale was found to be unidimensional. A similar procedure was followed for trait scales that had demonstrated multidimensionality in Study 1: two-factor (three for social intelligence) models were fit to this sample's data to constrain items (old and newly generated) to load exclusively onto their intended factor (Marsh, Lüdtke, Nagengast, Morin, & Von Davier, 2013). The lowest loading items were trimmed for each of the factors until adequate model fit was achieved. The process resulted in a total of 211 items that were found to be assessing 31 distinguishable character traits. Each character trait demonstrated strong internal consistency as shown in Table 4.

Exploration of character factor structure

Having established unidimensionality for each of the character trait scales, composite scores for each of the measures were created by calculating mean item scores. These scale-level character trait scores were used to assess the factor structure of character using exploratory structural equation modeling (ESEM; Asparouhov & Muthén, 2009). Individual differences measures (e.g. personality) have been found to conform poorly to CFA's restriction of zero cross-loadings and it has been proposed that ESEM is more appropriate (Hopwood & Donnellan, 2010). An important advantage of ESEM over traditional CFA is that ESEM allows for estimations of small, non-zero item factor loadings on factors for which those items were not originally intended to measure. Because the cross-loadings for each indicator will not be constrained to zero as they would be in CFA, correlations among latent factors will not be artificially-inflated, allowing for a more differentiated and clearer factor structure to emerge (Marsh et al., 2010).

Two criteria were used to assess number of factors to retain. First, parallel analysis involves generating multiple sets of random data with the same number of variables and observations as the data-set in question (Horn, 1965). Eigenvalues for components are generated across all these multiple sets of data and the factors from the original data-set whose eigenvalues exceeds 95% of the corresponding eigenvalues across the randomly generated datasets are recommended for retention. The second criterion used was the Bass-Ackwards approach using the recommended orthogonal varimax rotation (BA; Goldberg, 2006). This is essentially a series of orthogonal principal component analyses where component scores are correlated with those at adjacent levels to create path coefficients. This allows for examination of how components break down into 'smaller' components, which aids in their interpretation.

Parallel analysis indicated that a nine-factor solution would fit best. ESEM fit indices showed that an eight-factor solution fit the data most adequately (CFI = 0.96, TLI = 0.925, RMSEA = 0.06, SRMR = 0.02) when balancing parsimony and interpretability. The eight-component PCA structure from the BA approach corresponded with the eight-factor solution from the ESEM results fairly well; most of the character traits loaded onto the same emergent character core. To further explore the factor structure, the character traits corresponding to the eight-component solution yielded from the BA approach were used as the

Table 4. Character cores and underlying character traits.

Character core	Description	Character traits	Short description	Items	α
Appreciation	Savoring and valuing life	Gratitude	Noticing and seeking things to be thankful for	7	0.90
		Love	Valuing people in your life	4	0.82
Intellectual engagement	Proactive intellectual and emotional engagement with the world	Curiosity	Proactively seeking out a variety of interests and ideas	5	0.86
		Love of learning	Thirst for knowledge and skill acquisition for its own sake	10	0.93
		Creativity	Finding novel solutions and idea generation	4	0.88
		Appreciation of beauty	Finding beauty in everyday life	10	0.94
Fortitude	Determination marked by the optimism and energy needed to support staying power	Perseverance	Fierce determination in the face of obstacles	11	0.92
		Willpower	Persistence and commitment to important goals	7	0.89
		Zest	Enthusiasm and excitement about life	12	0.95
		Hope	Expecting the best	7	0.95
Interpersonal consideration	Capacity to get along with others well	Teamwork	Effort and sacrifice in attaining group aim	7	0.89
		Propriety	Acute sense of what the situation demands	4	0.82
		Social perceptiveness	Keen awareness of others' feelings and motivations	5	0.88
		Perspective	Giving good counsel rooted in experience	5	0.90
		Leadership	Uniting and motivating people towards a common goal	7	0.93
		Humor	Bringing laughter to people's lives	4	0.89
Sincerity	Resolute dependability and genuineness	Trustworthiness	Being true to one's word in deed	6	0.86
		Authenticity	Accurately presenting oneself regardless of the audience	6	0.85
		Bravery	Standing by convictions despite fear	5	0.87
		Humility	Avoiding attention- and praise-seeking	8	0.91
Temperance	Controlling short-term impulses to attain long-term goals	Carefulness	Being cautious in decision-making	4	0.81
		Forethought	Considering long-term consequences and all outcomes	4	0.80
		Self-control	Suppressing undesirable impulses	5	0.84
		Emotional awareness	Clear perception and understanding of one's emotions	4	0.85
Transcendence	Connection to something larger than the self	Spirituality	Seeking connection with a higher power	9	0.95
		Meaning/purpose	Sense that life fits into the grand scheme of things	10	0.96
Empathy	Balanced and benevolent discernment	Perspective-taking	Actively seeking understanding of contrary viewpoints	7	0.88
		Openness to evidence	Being unthreatened and receptive when confronted with evidence that contradicts one's beliefs	6	0.86
		Fairness	Desire to treat everyone justly and without bias	9	0.88
		Forgiveness	Letting past transgressions by others go so as to move forward	9	0.91
		Kindness	Finding joy in helping and relieving suffering of others	7	0.92

starting loading factors for a subsequent ESEM using target rotation (Browne, 2001). The resultant fit indices were identical to the previous ESEM (CFI = 0.96, TLI = 0.925, RMSEA = 0.06, SRMR = 0.02) as rotation does not change the overall fit.

Study 3: convergent/discriminant validity

In the present study, we first validated the unidimensionality of character traits and their factor structure to see if the dimensions that emerged in Study 2 could be recovered in a demographically diverse sample. Second, we sought to establish evidence for convergent validity by showing that our scales correlated relatively strongly with extant individual difference measures tapping the same or similar constructs as the character traits. Third, we sought to establish discriminant validity by showing that our scales did not correlate too strongly with potentially related constructs (i.e. personality, socially desirable responding, and moral cognitive development). Since the greatest current challenge in character trait research is empirically establishing their distinctness from what is already captured in the Big Five personality taxonomy (Noftle et al., 2011), we sought to test character traits' relative discriminability from Big Five personality facets. Additionally, we sought to address a second concern regarding the distinctness of character traits from engaging in impression management (Fowers, 2014), given that character traits are themselves socially desirable (Peterson & Seligman, 2004). We also aimed to examine whether character traits are distinct from well-established constructs within moral psychology, i.e. moral cognitive development. We hypothesized that character traits would be strongly associated with corresponding measures of the same or similar traits and moderately to weakly related with personality, social desirability responding, and moral cognitive development.

Method

Participants and procedure

Participants from MTurk (N = 1014; 45.3% male, 4 non-response; Age 18–78 years, M_{age} = 37.63, SD_{age} = 12.33, 1 non-response; White = 79.0%, Black = 8.9%, Asian/Pacific Islander = 5.7%, Hispanic or Latino = 4.6%, Native American/American Indian 0.3%, Other = 1.5%) were recruited from the United States to complete the study in exchange for $4.50. Workers were randomly assigned to one of three versions of the survey because of concerns about respondent fatigue. All three versions began with the character trait items presented in an item-randomized block and ended with a measure of social desirability, one item measuring political orientation (1 = very liberal to 5 = very conservative), a 10-item measure of social desirability responding (Fischer & Fick, 1993; Strahan & Gerbasi, 1972, $M = 4.78$, SD = 2.47, KR-20 = 0.74), and demographic questions. Version 1 also included a comprehensive facet-level measure of personality (IRT-based IPIP-120; Maples, Guan, Carter, & Miller, 2014) intended to tap the 30 facets of the NEO-PI-R (Costa, McCrae, & Dye, 1991). Of the 31 character traits assessed using the measure under development, a review of the individual differences literature revealed 24 extant individual difference measures that appeared to measure the same or similar constructs; version 2 and 3 each included a subset of these measures (see Appendix 1) and a short, 30-item facet-level personality measure that taps the NEO-PI-R facets: the *Five Factor Model-Rating Form* (FFMRF; Mullins-Sweatt, Jamerson, Samuel, Olson, & Widiger, 2006). Version 3 also included a measure of cognitive moral development, the *Defining Issues Test* (DIT; Rest, 1986) which presents six moral dilemmas and for each asks participants to rate the importance of twelve moral issues and then rank the top four most important issues. The N2 index score was calculated for this test, which takes into account both ranking as well as rating information to derive a score representing a greater degree of cognitive moral development or moral reasoning (Rest, Thoma, Narvaez, & Bebeau, 1997; $\alpha = 0.62$). N2 index scores were calculated for the 219 participants who took version 3 of the study survey and whose measure responses passed data screening criteria outlined in the DIT manual. Efforts were made to ensure test lengths of the three versions were comparable in item number.

Results

Unidimensionality of character traits

All character traits were re-analyzed for unidimensionality with Mplus 7.2 using maximum likelihood by way of CFA. All character traits evidenced generally good to acceptable fit statistics (Hu & Bentler, 1999). However, the propriety character trait (CFI = 0.94, TLI = 0.83, RMSEA = 0.24, 90% CI [0.20, 0.27], SRMR = 0.04) did not meet multiple cutoff criteria. Given the lack of replication, this scale was excluded from the measure under development.

Inspection of subsequent character trait scale score (i.e. scale item means) correlations indicated high intercorrelation between perseverance and willpower ($r = 0.91$), spirituality and meaning/purpose ($r = 0.87$), and curiosity and love of learning ($r = 0.86$). These correlated character trait pairs' respective items were analyzed using CFA and fit between one- and two-factor solutions compared using the chi-square difference test (Brown, 2015). Curiosity and love of learning demonstrated strong evidence of better fit using a two-factor solution with items loading onto their respective factors, $\Delta\chi^2(1) = 13.05$, $p < 0.001$, as did spirituality and meaning/purpose, $\Delta\chi^2(1) = 1936.50$, $p < 0.001$. However, fitting a two factor solution to the perseverance and willpower data did not lead to significantly better fit, $\Delta\chi^2(1) = 0.19$, $p = 0.28$, *ns*. As such, items for perseverance and willpower were grouped together and a scale score for this character trait (i.e. persistence) was calculated by averaging item scores. Removing the propriety character trait scale and combining willpower and perseverance into the persistence scale reduced the number of unique character traits from 31 to 29. Character trait correlations, reliabilities, and descriptives are presented in Appendix 2.

Factor structure of character

The factor structure of character was re-examined using ESEM with target rotation toward the structure that emerged in study 2. The eight-factor solution with target rotation demonstrated good overall fit (CFI = 0.97, TLI = 0.94, RMSEA = 0.05, 90% CI [0.05, 0.06], SRMR = 0.02). The pattern of loadings is generally as predicted although there were some moderate cross-loadings for appreciation of beauty, persistence, humility, and kindness character traits onto unpredicted character dimensions (see Table 5) and a character trait scale shift (i.e., teamwork from the character core of fortitude to interpersonal consideration). We move forward with our investigations using the measure we have refined to this point, which we call the Comprehensive Inventory of Virtuous Instantiations of Character (CIVIC).

Convergent and discriminant validity

To establish convergent validity, we examined correlations between each character trait and the most conceptually similar extant individual difference measure in the literature. We considered convergent validity established if the correlations were strong (i.e. above 0.50; Cohen, 1992). There is no hard cutoff for how high a correlation between two scale scores can be before lack of discriminant validity is implicated, but some have proposed a correlation of 0.85 (Brown, 2015; Clark & Watson, 1995) whereas other have suggested correlations higher than 0.90 (Kline, 2011) would indicate construct redundancy. We chose the lower cutoff of 0.85, a lower and easier hurdle to overcome to indicate lack of discriminant validity, since we did not have specific a priori hypotheses regarding which particular personality facet would be most highly correlated with each character trait. Instead, we sought to be comprehensive by examining correlations between each character trait

Table 5. Pattern matrix of character traits using ESEM (target rotation).

	Appreciation	Intellectual engagement	Fortitude	Interpersonal consideration	Sincerity	Temperance	Transcendence	Empathy
Gratitude	***0.47***		0.27		0.12		0.17	
Love	***0.50***		0.11		0.19			0.12
Curiosity		***0.99***						
Love of learning	−0.12	***0.90***						
Creativity		***0.47***		0.22	−0.14			
Appreciation of beauty	**0.30**	***0.42***					0.12	
Zest			***0.76***	0.11				
Hope	0.22		***0.75***		−0.12			
Persistence	−0.14		***0.49***	0.12	**0.38**	0.22		
Leadership	−0.13		0.22	***0.77***				
Social perceptiveness		0.11		***0.60***		0.15		
Perspective			−0.17	***0.52***		0.23		
Teamwork			0.15	***0.51***	0.14			0.24
Humor	0.15	0.12		***0.44***				
Trustworthiness			0.14		***0.65***	0.12		
Authenticity	0.15				***0.56***			0.11
Bravery		0.21		0.19	***0.50***			
Humility			−0.18	−0.18	***0.36***	0.13		**0.30**
Carefulness						***0.93***		
Forethought			0.14			***0.71***		
Self-control	−0.16		0.18		0.17	***0.49***		
Emotional awareness	0.23			0.28		*0.25*		0.11
Spirituality							***0.99***	
Meaning/purpose							***0.82***	
Perspective-taking								***0.90***
Forgiveness							0.15	***0.65***
Openness to evidence		0.17					−0.17	***0.69***
Fairness		0.12			0.27	0.11		***0.53***
Kindness	0.11			**0.35**	0.17	−0.12	0.18	***0.35***

Notes: Factor loadings with absolute value greater than or equal to 0.30 bolded. Factoring loadings with absolute value equal to or lower than 0.10 are suppressed. Loadings specified in target rotation in italics.

and all 30 personality facets to demonstrate that each character trait was discriminable from the full breadth of Big Five traits.

Table 6 displays correlations between each character trait with their corresponding extant individual difference tapping the same or similar construct, personality facets, social desirability responding, and moral cognitive development. For each character trait, correlations with the most similar extant individual difference measure tended to be strong, as hypothesized, although in some cases (character traits of perspective, teamwork, humor, humility, self-control) these correlations with similar constructs did not quite meet the cutoff of 0.50 to qualify as strong (rs range = 0.43–0.48). Nevertheless, the correlations were deemed high enough to suggest convergent validity.

In no case were character traits' correlations with personality facets so high as to suggest a lack of discriminant validity. Strongest correlations between each character trait and all personality facets ranged from 0.32 to 0.71, with most well below the cutoff of 0.85. These generally represented positive associations with personality facets, although for character traits of self-control, meaning/purpose, and spirituality the strongest personality facet correlate was negative. When considering each character traits' strong (i.e. $r > 0.50$) personality facet correlates, the domains of conscientiousness, agreeableness, and neuroticism (negatively) predominated in general.

Character traits also showed discriminability from, yet interpretable and differential relations with other constructs. Character traits tended to be weakly to moderately correlated with social desirability responding, largely mirroring how correlated the latter was with personality facets. Correlations between character traits and moral cognitive development were generally weak and non-significant. These results speak to character traits' discriminability from a relevant type of response bias and a well-established measure of moral cognitive development, respectively.

Discussion

Using a large demographically diverse MTurk sample, we established character traits' convergent validity with extant individual difference measures and discriminant validity from measures personality facets, social desirability, and moral cognitive development. Character traits generally

Table 6. Convergent and discriminant validity of character traits.

	Extant (target)	Personality[d]	Strongest personality facet correlate	SD[c]	DIT[e]
	r	Min. – Max. *r*		*r*	*r*
Gratitude	**0.76**[a]	−0.44 – *0.58*	Cheerfulness (E)	0.19	−0.11
Love	–	−0.33 – *0.52*	Altruism (A)	0.10	0.01
Curiosity	**0.61**[a]	−0.26 – *0.43*	Intellect (O)	0.05	0.12
Love of learning	**0.58**[a]	−0.27 – *0.45*	Intellect (O)	0.10	0.08
Creativity	**0.58**[a]	−0.36 – *0.43*	Intellect (O)	0.09	0.09
Appreciation of beauty	**0.72**[a]	−0.22 – *0.53*	Artistic interests (O)	0.15	0.11
Zest	**0.84**[b]	−0.68 – *0.74*	Cheerfulness (E)	0.25	−0.12
Hope	**0.71**[b]	−0.66 – 0.66	Cheerfulness (E) Depression (N)	0.17	−0.12
Persistence	**0.87**[b]	−0.55 – *0.70*	Self-efficacy (C)	0.24	−0.13
Leadership	**0.69**[b]	−0.52 – *0.68*	Assertiveness (E)	0.08	−0.09
Social perceptiveness	–	−0.27 – *0.36*	Self-efficacy (C)	0.07	−0.02
Perspective	0.43[b]	−0.42 – ***0.54***	Self-efficacy (C)	0.12	−0.03
Teamwork	0.45[b]	−0.31 – ***0.51***	Achievement striving (C)	0.19	−0.06
Humor	**0.48**[b]	−0.31 – *0.43*	Cheerfulness (E)	−0.04	−0.04
Trustworthiness	–	−0.40 – *0.66*	Dutifulness (C)	0.22	−0.11
Authenticity	**0.66**[a]	−0.43 – *0.55*	Dutifulness (C)	0.31	−0.03
Bravery	**0.53**[a]	−0.45 – *0.49*	Self-efficacy (C)	0.20	0.06
Humility	**0.47**[a]	−0.28 – *0.36*	Modesty (A)	0.23	−0.03
Carefulness	–	−0.31 – *0.51*	Caution (C)	0.13	0.01
Forethought	**0.58**[a]	−0.35 – *0.52*	Achievement striving (C)	0.15	−0.03
Self-control	**0.48**[a]	*−0.60* – 0.44	Immoderation (N)	0.22	0.00
Emotional awareness	**0.56**[a]	−0.37 – 0.45	Self-efficacy (C)	0.14	0.04
Spirituality	**0.87**[a]	*−0.49* – 0.29	Liberalism (O)	0.14	−0.18
Meaning/purpose	**0.83**[a]	*−0.48* – 0.43	Liberalism (O)	0.15	−0.24
Perspective-taking	**0.64**[b]	−0.30 – *0.37*	Altruism (A)	0.28	0.05
Forgiveness	**0.72**[b]	−0.26 – *0.41*	Altruism (A)	0.33	0.00
Fairness	–	−0.29 – *0.50*	Dutifulness (C)	0.23	0.17
Openness to evidence	–	−0.24 – *0.32*	Self-efficacy (C)	0.12	0.32
Kindness	**0.78**[b]	−0.22 – *0.70*	Altruism (A)	0.24	−0.04

Notes: Bolded numbers represent strongest correlation between character traits and other constructs. Italicized correlations indicate strongest magnitude of association between each character trait and personality facets. Extant (target) = extant individual difference scale tapping into same or similar construct as character trait. CIVIC = present study's measure. Extant (non-target) = all individual difference measures not tapping into similar construct as character trait. Personality = personality facets measured using the IRT-based IPIP-120 (Maples et al., 2014). SD = social desirability (Crowne & Marlowe, 1960; Strahan & Gerbasi, 1972), DIT = cognitive moral development measured using the N2 index of the Defining Issues Test (Rest, 1986).
[a]$n = 340$, version 3, $r = \pm 0.11$ is significant at $p < 0.05$.
[b]$n = 345$, version 2, $r = \pm 0.11$ is significant at $p < 0.05$.
[c]$n = 1014$, all versions, $r = \pm 0.07$ is significant at $p < 0.05$.
[d]$n = 329$, version 1, $r = \pm 0.11$ is significant at $p < 0.05$.
[e]$n = 219$, subset of version 3, $r = \pm 0.14$ is significant at $p < 0.05$.

correlated strongly with the target extant individual difference measure tapping a similar construct. In general, each character trait's strongest correlate was the extant individual difference measure thought to be the same or similar construct. Two exceptions were perspective and teamwork, both of which were slightly more associated with a personality facet.

Character traits' strongest personality facet correlates tended to be from the conscientiousness, agreeableness, and neuroticism domains. This is unsurprising given that personality researchers have recognized conscientiousness and agreeableness as the most evaluative, morally-salient dimensions of personality that respectively epitomize strength of will and goodness (McCrae & John, 1992). By extension, neuroticism could be argued to represent the antithesis of character traits that lead to well-being. Despite certain areas of high association between personality and character, the latter cannot be reduced to the former. For example, each of the three character traits of zest, hope, and persistence were strongly associated with conscientiousness, neuroticism (negative), and extraversion personality facets. That these patterns of correlations cut across personality domains indicates that the unidimensional character traits are not simply personality facets. Rather, they share latent structure space while mapping new areas and capturing meaningful, more evaluative individual difference variance.

Evidence suggests that what character captures is not simply individual differences in social desirability responding. Rather, character does not seem to be any more contaminated by socially desirable responding than personality. Personality's association with social desirability responding seems primarily driven by neuroticism facets like anger and anxiety. It is important to note that social desirability responding has two factors: self-deception (honest and overly positive response) and impression management (attempts at conveying a conventional and dependable persona) (Paulhus, 1991, but cf. Paulhus & John, 1998). The former is related to social adjustment and is at least partially substantive, whereas the latter is generally not related to social adjustment and should be controlled for in some cases (Paulhus, 1991). The social desirability scale used in the present study taps into both factors, although it mostly loads onto the impression management factor. Thus, it appears that neuroticism makes it difficult for people to convey a good impression, although doing so is not primarily an indication of substantive social adjustment. These results should allay some concerns regarding difficulty in obtaining accurate self-report of character due to self-report bias (Fowers, 2014).

Character traits also are not redundant with moral cognitive development, yet are able to capture salient dispositional tendencies thereof. That is, the two strongest positive character trait correlates of cognitive moral development were openness to evidence and fairness, which relate to unbiased consideration of opposing points of view and equitable treatment of others, respectively. They exemplify in traits the moral concern of justice that moral cognitive development was developed to assess (Kohlberg, 1969). The otherwise non-significant correlations between character traits and moral cognitive development mirrors Cawley et al.'s (2000) findings, perhaps as they suggest due to the different roots of study from which the emotional-motivational and intellectual-cognitive domains in psychology grew.

Study 4: Extending the nomological network

In this study, we sought to further situate character in a nomological net as well as delineate its distinctness from Big Five personality by examining their respective relationships with relevant constructs to demonstrate nomological validity (Campbell, 1960; Cronbach & Meehl, 1955). To address the former, we hypothesized that character would be positively associated with tendencies towards experiencing positive emotions and negatively with tendencies towards feeling negative emotions given the theoretical supposition and empirical evidence that character traits leads to greater well-being (*trans.* Irwin, 1999; Niemiec, 2013; Peterson & Seligman, 2004). We also hypothesized that character would be positively related to evaluative constructs that are rooted in the emotional-motivational domain (i.e. values, moral identity, and moral foundations).

To address the latter, we hypothesized that relative to personality, character would be less associated with constructs that are less salient to the moral domain (i.e. trait affectivity), but more highly associated with constructs more salient (i.e. values, moral identity, and moral foundations). Character is expected to relate to trait affectivity less than personality both because of the theory about its differential relation to *eudaimonia* vs. happiness or hedonic well-being (*trans.* Irwin, 1999) and evidence that affectivity is strongly related to personality (e.g. Weiss, Bates, & Luciano, 2008). However, character should be more related to values than personality because it has been argued that how values are prioritized guide how people conceptualize the best possible way of living, i.e. what will allow them to flourish as grounded in the Aristotelian conception of *eudaimonia* or thriving as opposed to what will lead to positive affect or happiness (Rohan, 2000). As such, it is expected that character will be more related to guides of what constitutes flourishing (i.e. values) because the cultivation of the former is theoretically a critical component of flourishing (*trans.* Irwin, 1999). Character was expected to relate more than personality to moral identity because moral identity, as currently assessed, is operationalized as the degree that traits typical to lay conceptions of moral character are important to a sense of self. Since most of the these traits are directly assessed in the CIVIC, it is reasonable to expect that feeling these traits are important to one's identity will relate more strongly to levels of these specific traits than associated, but distinguishable traits in personality. Finally, character is hypothesized to be more related than personality to moral foundations because the virtues that describe character have been judged by religions and philosophies as morally good (Dahlsgaard et al., 2005). Thus, character should be more related to intuitions about what is morally good than descriptive accounts of traits (i.e. personality).

Method

Participants and procedures

Participants were 200 undergraduate student sample recruited from a Midwestern university via an online platform (49.5% male, 4; ages 18–36, $M_{age} = 19.64$, $SD_{age} = 1.65$; White = 73.5%, Asian = 16.5%, Latin American = 4.0%, Black = 3.5%, Other = 2.5%). Participants completed the study online in exchange for two research credits for participation. Character was measured using the CIVIC and personality measured as in Study 3 using the FFMRF (Neuroticism $M = 2.57$, $SD = 0.69$, $\alpha = 0.78$; Extraversion $M = 3.51$, $SD = 0.63$, $\alpha = 0.76$; Openness to Experience $M = 3.5$, $SD = 0.59$, $\alpha = 0.70$; Agreeableness $M = 3.72$, $SD = 0.56$, $\alpha = 0.69$; Conscientiousness $M = 3.81$, $SD = 0.54$, $\alpha = 0.76$). Several measures were also included to test our hypotheses regarding the nomological validity of character (Cronbach & Meehl, 1955). We sought first to distinguish character from dispositional affectivity using the *Positive and Negative Affect Schedule* (PANAS; Watson, Clark, & Tellegen, 1988), which taps into tendencies to experience positive and negative moods. We then focused on comparing character and personality's associations with moral constructs. First, moral identity was assessed using the *Self Importance of*

Moral Identity Scale developed by Aquino and Reed (SIMI; 2002). The SIMI presents nine words (i.e. *caring*, *compassionate*, *fair*, *friendly*, *generous*, *hardworking*, *helpful*, *honest*, and *kind*) generated and selected from lay conceptions of moral character and then asks respondents to answer 10 items on a 1 (strongly disagree) to 5 (strongly agree) scale to assess how central these characteristics are to their sense of self (internalization) and how fully they are represented in external markers of behavior (symbolization). Second, moral intuitions about what is right or wrong was measured using the *Moral Foundations Questionnaire* (MFQ; Graham et al., 2011). The MFQ assesses which and to what degree five posited moral concerns form the basis for a person's moral judgment (Haidt & Graham, 2007). The foundation of harm/care relates to concerns about the distress and suffering of others. The fairness/justice foundation relates to issues about reciprocity and more abstract principles of equity. The ingroup/loyalty foundation relates to issues surrounding patriotism, sacrifice for the good of the group, and group allegiance. The authority/respect foundation relates to concerns about deference to authorities and role fulfillment in hierarchies. The purity/sanctity foundation relates to concerns regarding physical or social contamination. Finally, how participants prioritize their values was assessed using the *Schwartz Values Survey* (SVS-57; Schwartz, 1992). The measure taps into 10 distinct value types that can be modeled as a value system circumplex indicating the degree of similarity between types. Those adjacent in the circumplex are more similar, whereas those farther away are increasingly dissimilar to the point where opposing value types are in tension or even incompatible with each other (Maio, Pakizeh, Cheung, & Rees, 2009). The internal consistencies of the value types are somewhat varied (see Table 7), but are comparable to those found in other studies (see Sagiv & Schwartz, 2000) and are expected given both low numbers of items to assess each value type as well as their relative broadness (Schwartz et al., 2001). Scores were recalculated for correlation analysis using procedures provided by Schwartz (2013) to account for scale use tendencies.

The 10 value types in the circumplex further describe two underlying motivational dimensions: self-enhancement (i.e. achievement and power value types) vs. self-transcendence (i.e. universalism and benevolence value types) and openness to change (i.e. hedonism, stimulation, and self-direction value types) vs. conservation (i.e. security, conformity, and tradition value types), the scores of which are calculated by averaging their respective value types scores.

Results

Preliminary analyses

We decided to aggregate the character traits to the character core level to align the breadth of the character constructs with those of personality domains and the other constructs in the current study. Prior to doing so, we tested whether the factor structure of character traits revealed in Study 3 using an MTurk sample would also emerge in this undergraduate sample using ESEM with target rotation. The model demonstrated acceptable overall fit (CFI = 0.95, TLI = 0.90, RMSEA = 0.07, 90% CI [0.06, 0.08], SRMR = 0.03), although there was a scale shift (i.e., humility loaded on the character core of empathy instead of sincerity). Scale items for each scale were averaged to obtain scale scores and scale scores averaged to create composite character core scores.

Character and nomological validity

Table 7 displays descriptives for the study variables. We sought to establish first that character (as a set of character cores) were significantly related to the nomological validity scales by regressing the latter on the former using multiple regression (Cohen, Cohen, West, & Aiken, 2002). Then we explored how character and personality are situated in a nomological net by examining how they differentially related to the nomological validity scales using partial correlations (Kern, Waters, Adler, & White, 2015) given intercorrelations between all three sets. Each character core within the character set was correlated with the nomological validity scales while partialling out the variance in each shared with all Big Five personality domains. Similarly, each personality domain was correlated with the nomological validity scales while partialling out the variance in each shared with all eight character cores. These partial correlations demonstrate how different each character core is from the entire set of personality domains (and vice versa) in how the former relates to the related scales. Finally, we conducted relative weight analysis (RWA; Johnson, 2000; see Tonidandel & LeBreton, 2011) to provide evidence for our hypothesis that character would be more associated with more morality-related constructs, but personality would be more highly related to positive, but morally neutral constructs. RWA is a particularly well-suited technique for the present hypothesis since it partitions the amount of outcome variance attributable to the appropriate predictors even when they are multicollinear, which allows for direct comparisons in the explanatory power of predictors.

Figure 1 shows that character as a set of character cores generally explains a significant amount of variance in each nomological validity scale, the one exception being the moral foundation of fairness/justice. Table 8 presents the results of partial correlation analyses between character cores and personality domains with the related constructs. Personality had the single highest partial correlation with the trait affectivity scales. The more evaluative constructs (i.e. moral identity, moral foundations, and values) had the

Table 7. Character and personality descriptives and zero-order correlations with nomological validity scales.

	M	SD	*α*	App	IE	F	IC	S	Te	Tr	Em	N	E	O	A	C
Character																
Appreciation	3.49	0.36	0.70													
Intellectual engagement	3.10	0.40	0.77	0.44												
Fortitude	3.08	0.44	0.86	0.56	0.56											
Interpersonal consideration	3.13	0.36	0.79	0.50	0.59	0.62										
Sincerity	3.22	0.32	0.69	0.57	0.51	0.59	0.60									
Temperance	3.12	0.37	0.80	0.48	0.53	0.52	0.60	0.61								
Transcendence	2.79	0.70	0.89	0.41	0.18	0.49	0.41	0.33	0.28							
Empathy	3.17	0.30	0.81	0.51	0.58	0.48	0.58	0.64	0.55	0.22						
Personality																
Neuroticism	2.57	0.69	0.78	−0.25	−0.25	−0.40	−0.29	−0.27	−0.25	−0.04	−0.21					
Extraversion	3.52	0.63	0.76	0.41	0.28	0.55	0.54	0.22	0.13	0.42	0.20	−0.31				
Openness	3.51	0.59	0.70	0.24	0.47	0.21	0.35	0.26	0.19	0.01	0.32	−0.11	0.33			
Agreeableness	3.72	0.56	0.69	0.36	0.15	0.33	0.31	0.36	0.26	0.35	0.35	−0.08	0.31	0.22		
Conscientiousness	3.82	0.54	0.76	0.36	0.28	0.50	0.34	0.38	0.48	0.30	0.21	−0.16	0.20	0.00	0.40	
Trait affectivity																
Negative affectivity	2.16	0.62	0.86	**−0.20**	−0.12	**−0.29**	−0.13	**−0.22**	−0.07	−0.04	−0.10	0.52	**−0.23**	−0.02	−0.05	0.03
Positive affectivity	3.55	0.61	0.88	**0.46**	**0.49**	0.69	**0.54**	**0.41**	**0.37**	**0.33**	**0.33**	**−0.43**	**0.65**	**0.30**	**0.38**	**0.48**
Moral identity																
Internalization	4.44	0.49	0.69	0.40	0.10	**0.29**	**0.30**	**0.32**	**0.20**	**0.33**	**0.29**	−0.13	**0.24**	−0.02	**0.31**	**0.30**
Symbolization	3.61	0.57	0.75	**0.33**	**0.27**	**0.44**	**0.40**	**0.28**	**0.21**	0.49	**0.29**	−0.02	**0.38**	**0.17**	**0.40**	**0.24**
Moral foundations																
Harm	4.52	0.67	0.58	**0.31**	**0.16**	**0.26**	**0.24**	**0.26**	**0.23**	**0.25**	**0.29**	−0.12	**0.22**	**0.21**	0.35	**0.25**
Fairness	4.53	0.55	0.60	**0.20**	0.11	0.11	**0.16**	**0.18**	**0.24**	0.05	**0.15**	0.00	0.14	0.27	**0.18**	**0.17**
Ingroup	4.08	0.77	0.68	0.13	0.01	**0.35**	**0.24**	**0.15**	**0.19**	0.44	0.03	−0.14	**0.44**	0.01	**0.27**	**0.26**
Authority	3.97	0.71	0.64	0.08	−0.08	**0.23**	0.11	0.10	0.11	0.40	0.02	0.00	**0.23**	**−0.17**	**0.24**	**0.25**
Purity	3.67	0.89	0.70	**0.15**	**−0.10**	**0.26**	0.08	**0.17**	0.13	0.59	−0.02	0.08	**0.20**	−0.13	**0.32**	**0.32**
Progressivism	0.62	0.67	0.77	0.09	0.19	−0.16	0.02	0.04	0.05	−0.42	**0.19**	−0.04	**−0.17**	**0.33**	−0.08	**−0.13**
Values																
Achievement	4.58	0.98	0.64	0.03	−0.02	0.13	0.12	0.04	0.00	−0.19	−0.05	−0.07	0.05	−0.07	**−0.17**	0.10
Power	2.59	10.42	0.76	**−0.32**	**−0.30**	**−0.19**	−0.06	−0.41	**−0.19**	**−0.16**	**−0.30**	**0.15**	0.06	−0.07	**−0.19**	−0.08
Security	4.17	0.96	0.60	−0.14	−0.26	−0.04	**−0.16**	−0.13	−0.05	−0.09	**−0.22**	0.11	−0.09	**−0.15**	−0.04	0.10
Tradition	3.45	1.11	0.54	0.08	**−0.17**	**0.17**	0.05	0.13	0.07	0.51	0.05	0.03	0.07	**−0.39**	**0.29**	0.13
Conformity	4.47	1.00	0.70	0.13	**−0.15**	**0.17**	0.01	**0.16**	0.07	0.28	0.05	−0.03	0.01	**−0.19**	**0.26**	**0.17**
Benevolence	5.05	0.79	0.59	0.15	−0.12	−0.07	−0.03	0.24	0.06	0.06	0.12	−0.01	−0.13	**−0.16**	**0.15**	0.06
Universalism	4.26	0.97	0.78	0.09	0.34	−0.12	0.04	0.14	0.11	−0.17	**0.28**	−0.03	−0.14	**0.31**	0.05	−0.13
Self-direction	4.62	0.91	0.65	−0.13	0.38	−0.05	−0.01	0.03	−0.02	**−0.37**	0.02	−0.08	−0.09	**0.33**	**−0.29**	−0.12
Stimulation	3.96	1.20	0.68	−0.15	0.24	0.07	0.08	−0.08	−0.11	−0.15	−0.04	−0.12	**0.17**	**0.22**	**−0.15**	**−0.16**
Hedonism	4.42	1.13	0.61	**−0.16**	−0.09	−0.13	−0.12	**−0.18**	−0.14	−0.38	−0.11	0.01	0.00	**0.16**	**−0.24**	**−0.21**
Self enhancement	3.59	1.05	0.66	**−0.27**	**−0.25**	−0.08	0.01	−0.30	**−0.15**	−0.22	**−0.26**	0.08	0.07	−0.09	**−0.23**	−0.01
Conservation	4.03	0.85	0.78	0.05	**−0.29**	**0.17**	−0.04	0.09	0.05	0.39	−0.04	0.05	0.01	**−0.38**	**0.28**	**0.20**
Self-transcendence	4.66	0.72	0.49	**0.17**	**0.18**	−0.14	0.01	**0.28**	0.13	−0.09	0.31	−0.03	**−0.20**	0.13	**0.15**	−0.06
Openness to change	4.33	0.88	0.74	**−0.21**	**0.23**	−0.05	−0.02	−0.12	−0.14	−0.42	−0.07	−0.09	0.06	**0.33**	**−0.32**	**−0.24**

Notes: App = appreciation, IE = intellectual engagement, F = fortitude, IC = interpersonal consideration, S = sincerity, Te = temperance, Tr = transcendence, Em = empathy, N = neuroticism, E = extraversion, O = openness to experience, A = agreeableness, and C = conscientiousness. All significant correlations ($p < 0.05$) shown in boldface. Strongest correlation in row underlined.

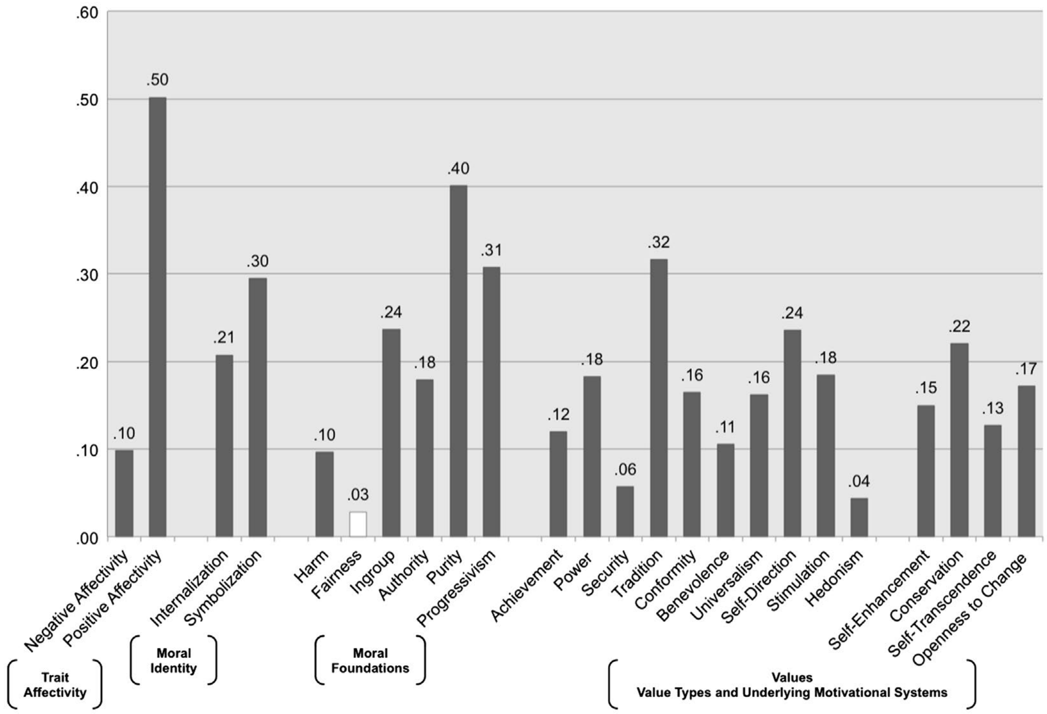

Figure 1. Variance explained (adjusted R^2) in nomological validity constructs by character.
Note: White bars indicate non-significant variance explained ($p > 0.05$).

Table 8. Partial correlations of character cores (partialiing out personality) and personality domains (partialling out character) with nomological validity scales.

	App	IE	F	IC	S	Te	Tr	Em	N	E	O	A	C
Trait affectivity													
Negative affectivity	−0.11	−0.05	**−0.17**	0.02	**−0.17**	−0.01	0.00	−0.02	0.45	−0.13	0.05	0.04	***0.21***
Positive affectivity	0.10	***0.28***	***0.32***	0.12	0.14	0.11	0.02	0.12	***−0.22***	0.42	**0.14**	***0.26***	***0.26***
Moral identity													
Internalization	0.28	0.03	0.08	**0.16**	***0.20***	0.07	**0.17**	***0.22***	−0.05	0.01	−0.12	0.11	***0.20***
Symbolization	0.13	***0.20***	***0.30***	***0.22***	**0.15**	0.11	0.33	**0.18**	**0.14**	0.05	0.07	***0.24***	0.06
Moral foundations													
Harm	0.14	−0.01	0.05	0.05	0.08	0.08	0.13	0.14	−0.04	0.09	**0.17**	0.21	0.13
Fairness	0.08	−0.07	−0.03	0.01	0.06	0.15	−0.03	0.03	0.08	0.15	0.26	0.12	0.07
Ingroup	−0.13	−0.12	0.07	−0.02	0.01	0.11	***0.25***	−0.09	−0.08	0.33	0.07	0.15	0.07
Authority	−0.07	−0.08	0.08	−0.01	0.03	0.06	0.25	−0.01	0.03	0.13	−0.11	0.13	0.15
Purity	0.00	−0.14	0.14	−0.07	0.08	0.05	0.48	−0.09	0.12	0.07	0.00	**0.19**	***0.21***
Progressivism	**0.18**	0.10	−0.10	0.07	0.02	0.03	−0.34	0.17	−0.01	−0.09	***0.24***	−0.01	−0.06
Values													
Achievement	−0.02	0.05	0.18	0.13	0.02	0.03	−0.11	0.01	0.09	0.00	−0.06	−0.01	0.14
Power	***−0.26***	**−0.20**	−0.10	0.00	−0.32	−0.06	−0.12	**−0.18**	0.13	0.11	0.07	0.03	0.11
Security	−0.07	−0.08	0.09	−0.06	−0.11	0.01	−0.02	−0.12	0.14	0.02	0.02	0.13	0.16
Tradition	0.06	0.09	***0.26***	0.11	0.11	**0.16**	0.44	0.09	0.05	−0.01	***−0.26***	***0.23***	0.09
Conformity	0.08	0.00	***0.22***	0.03	0.06	0.08	0.23	0.03	0.05	−0.02	−0.06	***0.22***	0.13
Benevolence	**0.19**	0.07	0.07	0.08	0.20	0.12	0.14	0.14	0.01	0.02	−0.10	**0.18**	**0.15**
Universalism	0.12	0.33	0.06	0.09	0.07	**0.14**	0.02	**0.19**	0.01	0.05	0.13	**0.17**	0.00
Self-direction	−0.06	0.33	0.11	0.03	0.01	0.02	−0.11	0.03	0.02	0.08	**0.15**	−0.01	0.03
Stimulation	**−0.16**	0.23	0.12	0.03	−0.09	−0.04	−0.06	−0.05	−0.03	**0.17**	0.12	0.04	−0.07
Hedonism	−0.07	−0.05	0.03	−0.06	−0.12	−0.01	−0.21	−0.05	0.07	**0.15**	**0.18**	−0.01	−0.05
Self enhancement	**−0.18**	−0.11	0.01	0.06	−0.21	−0.03	−0.13	−0.12	0.13	0.08	0.02	0.02	0.14
Conservation	0.03	0.00	***0.24***	0.03	0.03	0.10	0.27	0.00	0.10	0.00	−0.12	***0.24***	**0.16**
Self-transcendence	**0.18**	0.26	0.08	0.10	**0.16**	**0.16**	0.09	***0.21***	0.01	0.04	0.03	***0.21***	0.09
Openness to change	−0.13	0.20	0.11	0.00	−0.09	−0.01	**−0.16**	−0.03	0.03	**0.17**	**0.18**	0.01	−0.04

Notes: App = appreciation, IE = intellectual engagement, F = fortitude, IC = interpersonal consideration, S = sincerity, Te = temperance, Tr = transcendence, Em = empathy, N = neuroticism, E = extraversion, O = openness to experience, A = agreeableness, and C = conscientiousness. All significant correlations $p < 0.05$ shown in boldface and those $p < 0.01$ also italicized. Strongest correlation in row underlined.

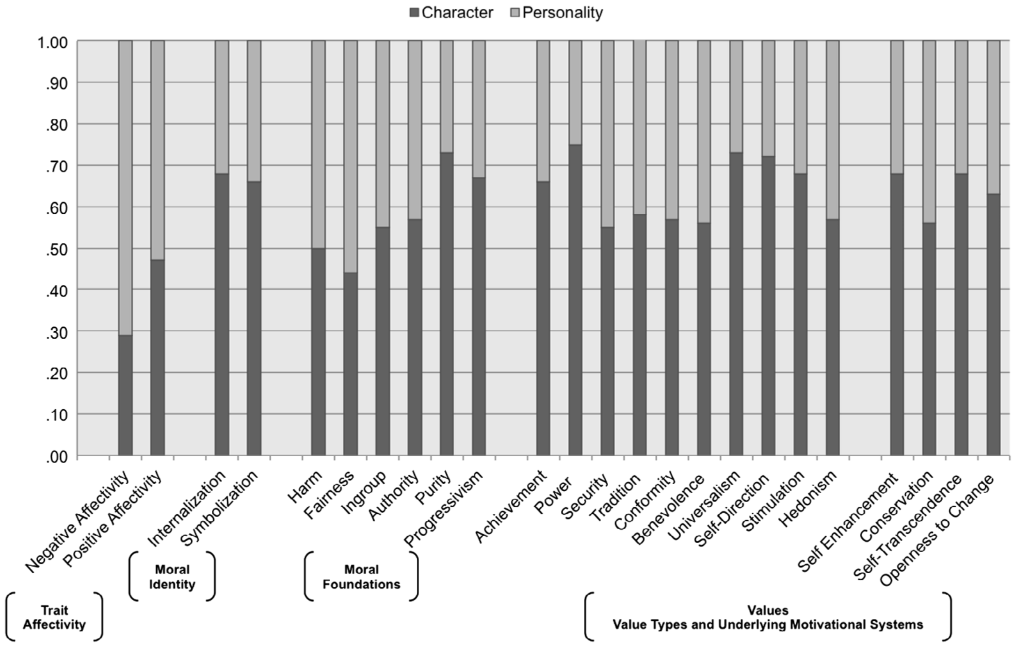

Figure 2. Proportion of explained variance in nomological validity constructs attributable to character vs. Big Five personality.

single highest partial correlation with character cores in general, although the moral foundations of harm/care, fairness/justice, and ingroup/loyalty actually had higher partial correlations with personality domains. As can be seen in Figure 2, the percentage of explained variance in the morality-salient measures that is attributed to character cores is generally higher compared to personality, although personality explained slightly more variance in the fairness foundation and both explained approximately equal variance in the harm foundation. In contrast, personality explained a large proportion of variance in negative affectivity and slightly more variance in positive affectivity compared to character. This is encouraging evidence for character's status as a morally-salient construct.

Discussion

We found evidence that as expected character was associated with an array of relevant constructs, and also in ways meaningfully distinguishable from personality. Of particular note is the character core of transcendence, a multidimensional individual difference factor that is not subsumed by the Big Five (Piedmont, 1999) and seems to capture trait manifestations of moral foundations that have been relatively understudied and underrepresented within moral psychology (i.e. ingroup/loyalty, authority/respect, and purity/sanctity or the binding foundations; Graham et al., 2011). Our hypothesis that character would be relatively more associated with morality-related constructs and personality relatively more associated morally-neutral constructs was also generally supported. In sum, the study demonstrated that character is distinguishable from personality in the way it relates to other constructs, establishing nomological validity.

Study 5: Test–retest reliability and predictive validity

Virtues have long been conceived of as human excellences that engender performance and flourishing for self and society (Sandage & Hill, 2001). Consequently, confirming the ability of character cores to predict such outcomes is therefore a particularly critical step in providing empirical support for construct validity. The primary purpose of this study was to explore relationships between character cores and various measures of objective criteria of performance, social adjustment, and well-being. We expected that character as a set would explain significant variance in all outcomes. We sought also to compare the relative explanatory power and incremental validity of character cores compared to personality in explaining these outcomes. We expected character to increment beyond personality in all outcomes given that these outcomes are those that good character should engender. A secondary aim is to examine the stability of character cores over time. Test–retest reliability is the most commonly used index of trait stability in constructs such as personality (Caspi & Roberts, 1990). High test–retest reliability would add support for the use of the measure as a reliable measurement of character cores as is currently conceptualized.

Method

Participants and procedures

MTurk participants from Study 3 and undergraduate student participants from Study 4 were contacted to participate in the current study. The former were messaged via an R statistical software package and script (Wiegand, 2015) that allowed emails to be sent to the email address associated with their respective MTurk worker ID and the latter were mass emailed through an online survey platform emailing system. MTurk participants completed the current study's survey 50–63 days (M = 54.18, SD = 3.23) after they completed the Study 3 survey in exchange for $4.00. The interval between completing the Study 4 survey and the current study's survey was markedly shorter for the undergraduate sample (15–30 days, M = 22.39, SD = 3.04); this sample received two research credits for participation.

Data from 486 Mturk participants (42.6% male; Age 18–78 years, M_{age} = 39.47, SD_{age} = 12.61; White = 82.1%, Black = 8.2%, Asian = 6.0%, Latin American = 2.3%, Other = 1.4%) and 41 undergraduate students (58.5% male; Age 18–23 years, M_{age} = 19.61, SD_{age} = 12.22; White = 78.0%, Black = 2.4%, Asian = 14.6%, Latin American = 2.4%, Other = 2.4%) were retained for analysis after data cleaning. This included conventional cleaning procedures (e.g. removing those who failed quality control checks) as well as screening out participants who indicated through a comment box some unique life event (e.g. recently diagnosed illness in a close relative) had affected their ratings of criteria measures (N = 2).

Participants were presented the CIVIC in an item-randomized block as well as the FFMRF to measure Big Five personality. Performance in life was assessed and operationalized as measures of outcomes across the life domains of health, work, and love. Specifically, health was assessed as five items from the *Health-Related Quality of Life Scales* (HRQoL; Centers for Disease Control and Prevention [CDC], 2011). Four of these items that inquire how many days in the past 30 days respondents have experienced certain general physical symptoms load onto a physical health factor (Mielenz, Jackson, Currey, DeVellis, & Callahan, 2006). The three items that assess negative health outcomes (e.g. pain that prevented normal activities) were reverse scored and these four items were averaged to attain an overall index of physical health (M = 25.87, SD = 4.59, α = 0.68). Additionally, one item from the healthy days core module that also loaded onto the physical health factor (Mielenz et al., 2006) which assesses general health on a 5-point Likert scale ranging from poor to excellent was used separately to avoid scaling issues. In the work domain, the performance outcome for students was operationalized as their grade point average (GPA) at the end of the spring 2015 semester as obtained through the Office of the Registrar for students who agreed to release this information when asked at either Time 1 or Time 2. For MTurk participants, performance was operationalized as level of educational attainment and personal individual gross income. The former was assessed on a 1 = (Grade 1–6) to 14 = Doctoral degree (in academic graduate school, e.g. Ph.D. or Ed.D.) (M = 9.15, SD = 4.48) scale and the latter on an scale from 1 = ($0–$19,999) to 11 = ($200,000+) increasing in $20,000 increments ($M$ = 2.32, SD = 1.37). Finally, positive outcomes in the love domain were assessed using *The Multidimensional Scale of Perceived Social Support* (Zimet, Dahlem, Zimet, & Farley, 1988), a 12-item scale that measures perceptions about how sufficient social support is in a person's life on a scale from 1 (very strongly disagree) to 5 (very strongly agree). The overall measure (M = 4.09, SD = 0.73, α = 0.93) has three subscales that measure social support received from family (M = 4.05, SD = 0.92, α = 0.94), significant others (M = 4.17, SD = 1.00, α = 0.98), and friends (M = 4.06, SD = 0.78, α = 0.92).

Social adjustment was measured in two ways. First, the eight-item responsibility subscale of the self-restraint dimension of the *Weinberger Adjustment Inventory* is measured on a 1 = (False) to 5 = (True) scale (WAI-RES; Weinberger & Schwartz, 1990; M = 4.60, SD = 0.46, α = 0.82) and taps into the tendency to restrain egoistic drives that violate broad social norms (e.g. 'I do things that are against the law more often than most people' [reverse scored]). Second, the *Prosocial Tendencies Measure* (Carlo & Randall, 2002; M = 3.17, SD = 0.51, α = 0.84) measures six related, but distinct aspects of prosocial tendencies with theoretically different motivational bases and characteristics. The public dimension measures the tendency to act prosocially in part to gain social approval and esteem (M = 1.58, SD = 0.80, α = 0.83). In contrast, the anonymous subscale measures the tendency to prefer to act prosocially without being noticed or seen engaging in the act (M = 3.10, SD = 1.00, α = 0.88). The compliant subscale assessses prosociality rooted more in approval-seeking (M = 3.60, SD = 0.97, α = 0.86 [two items]). The emotional subscale taps into responsiveness to emotionally evocative situations (e.g. a baby crying; M = 3.13, SD = 0.96, α = 0.84) and relatedly the dire subscale taps into tendencies toward responsiveness to crises (M = 3.24, SD = 0.94, α = 0.80). Lastly, the altruism subscale assesses prosocial tendencies grounded in a sense of empathy and internalized rules or norms (M = 4.37, SD = 0.74, α = 0.80).

Finally, we endeavored to assess a full conception of mental well-being starting with indices of ill-being. We used the *Generalized Anxiety Disorder-7* (GAD-7; Spitzer, Kroenke, Williams, & Löwe, 2006; M = 1.51, SD = 0.59, α = 0.90) to assess frequency of anxiety symptomology and the *Patient Health Questionnaire* as a brief, nine-item measure to assess how frequently respondents have experienced symptomology associated with depression (PHQ-9; Kroenke, Spitzer, & Williams, 2001; $M = 1.47$, SD = 0.52, $\alpha = 0.88$). Both scales use response options ranging from 0 (not at all) to 3 (nearly every day). We then assessed the degree to which psychological needs are met. Within self-determination theory, human beings are thought to require three basic psychological needs for mental health and optimal performance: autonomy ($M = 5.25$, SD = 0.93, $\alpha = 0.79$), competence ($M = 5.24$, SD = 1.03, $\alpha = 0.78$), and relatedness ($M = 5.27$, SD = 0.97, $\alpha = 0.83$) (Basic Psychological Needs Scale; Deci & Ryan, 2000). Response options range from 1 (not at all true) to 7 (very true). Moving towards assessment of well-being, subjective well-being was calculated by administering a measure of general positive and negative affect (PANAS; Watson et al., 1988), subtracting the negative affect scores from the positive affect scores, and then adding this difference to scores on the *Satisfaction with Life Scale* (Diener, Emmons, Larsen, & Griffin, 1985) as has been done in prior studies (e.g. Brunstein, 1993; $M = 42.71$, SD = 15.94, $\alpha = 0.83$). Finally, the Psychological Well-Being scales were administered (PWB; Ryff, 1989) to tap into broader, more inclusive conceptions of well-being beyond hedonic well-being that arguably constitutes thriving or flourishing (Waterman, 1993). The PWB scales are comprised of six, 14-item subscales measuring various dimensions of positive psychological functioning, each on a 1 (strongly disagree) to 6 (strongly agree) scale. These dimensions include unconditional self-acceptance ($M = 4.25$, SD = 1.05, $\alpha = 0.94$), autonomy or self-reliance for approval ($M = 4.57$, SD = 0.77, $\alpha = 0.88$), environmental mastery or the ability to choose one's environment or shape an existing one into one that is suitable ($M = 4.41$, SD = 0.90, $\alpha = 0.92$), personal growth towards one's potential ($M = 4.77$, SD = 0.67, $\alpha = 0.87$), having a sense of purpose or meaning in life ($M = 4.53$, SD = 0.90, $\alpha = 0.92$), and positive relationships with others marked by intimacy ($M = 4.52$, SD = 0.91, $\alpha = 0.92$).

Results

Test–retest reliability

Test–retest reliabilities across both samples were high ($M = 0.77$; Range = 0.65 [carefulness] to 0.92 [spirituality]), although MTurkers tended to have higher test–retest reliability ($M = 0.77$; Range = 0.65 [carefulness] to 0.92 [spirituality]) compared to undergraduates ($M = 0.68$; Range = 0.39 [forethought] to 0.85 [spirituality]) despite a longer time interval between assessments.

Predictive and incremental validity

Time 1 character trait scores were aggregated to the character core level, again to match the breadth of the comparison construct (Time 1 personality domains). A comparison of overall predictive validity between character cores and personality domains was examined using RWA. We ran RWA using the character and personality scores measured at Time 1 as predictors of outcomes measured at Time 2. Character's and personality's incremental validity beyond each other in explaining each outcome was assessed using hierarchical regression analyses (Cohen et al., 2002).

Character explained significant incremental variance beyond personality in all outcomes aside from income, establishing incremental validity (see Figure 3). In the majority of cases the incremental variance in outcomes explained by character was larger than that explained by personality, exceptions being income, GPA, self-restraint, anxiety, and depression. A similar pattern was found with the results from the RWA: character as a set explained the majority of variance in all criterion measures compared to personality except for the five aforementioned outcomes (see Figure 4).

Discussion

The results were generally as hypothesized. The test–retest reliabilities were in most cases reasonably high and comparable to those of personality domains. Personality provided better prediction of certain performance metrics (i.e. personal income) and measures of mental ill-being such as depressive and anxiety-related symptomology. These results were largely due to the detrimental effects of being high on neuroticism. Across all other measures of performance, mental well-being, and most measures of societal adjustment, character cores explained as much or significantly more variance than personality domains. In sum, the results demonstrate the importance of character cores in predicting performance, social, and well-being outcomes.

General discussion

Summary of findings

Using an iterative process of item selection and refinement, unidimensional character traits were delineated in Studies 1 and 2 that proved critical in revealing a coherent factor structure. Contrary to other empirical investigations using the VIA-IS, the factor structure that emerged from empirical analysis of the CIVIC using an MTurk sample revealed eight overarching character cores that collectively describe good character and, moreover, replicated across both an MTurk and undergraduate sample, demonstrating generally acceptable to good fit (all CFIs ≥ 0.95, TLIs ≥ 0.90, RMSEAs ≤ 0.07, and SRMRs ≤ 0.03). As a point of comparison, this was similar or better fit compared to ESEM fit indices for the five factor model of personality using 60-item

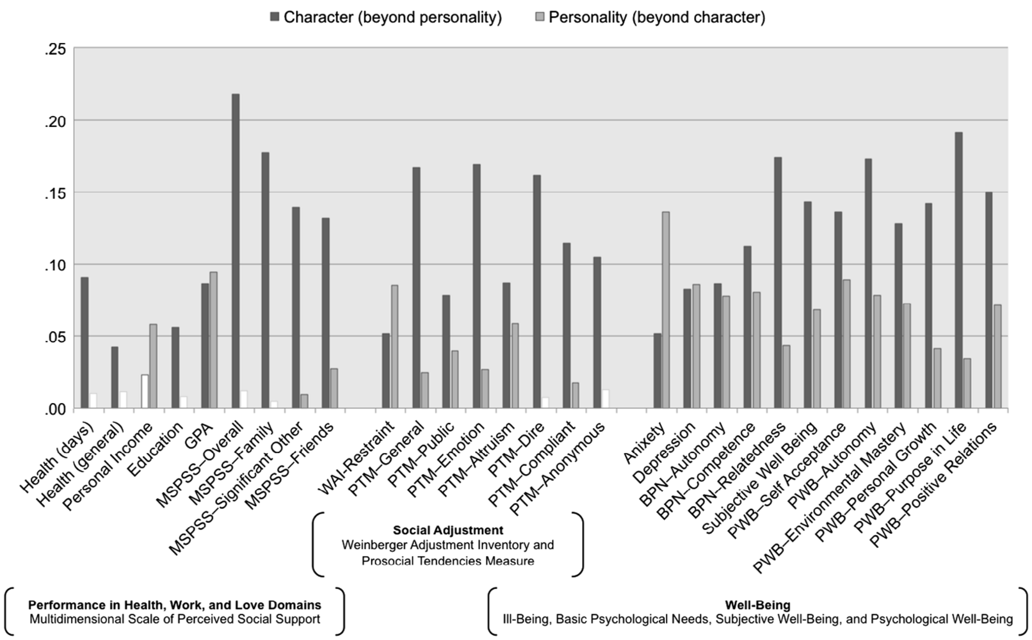

Figure 3. Incremental variance in criteria explained by character and Big Five personality.
Note: White bars indicate non-significant incremental variance explained ($p > .05$).

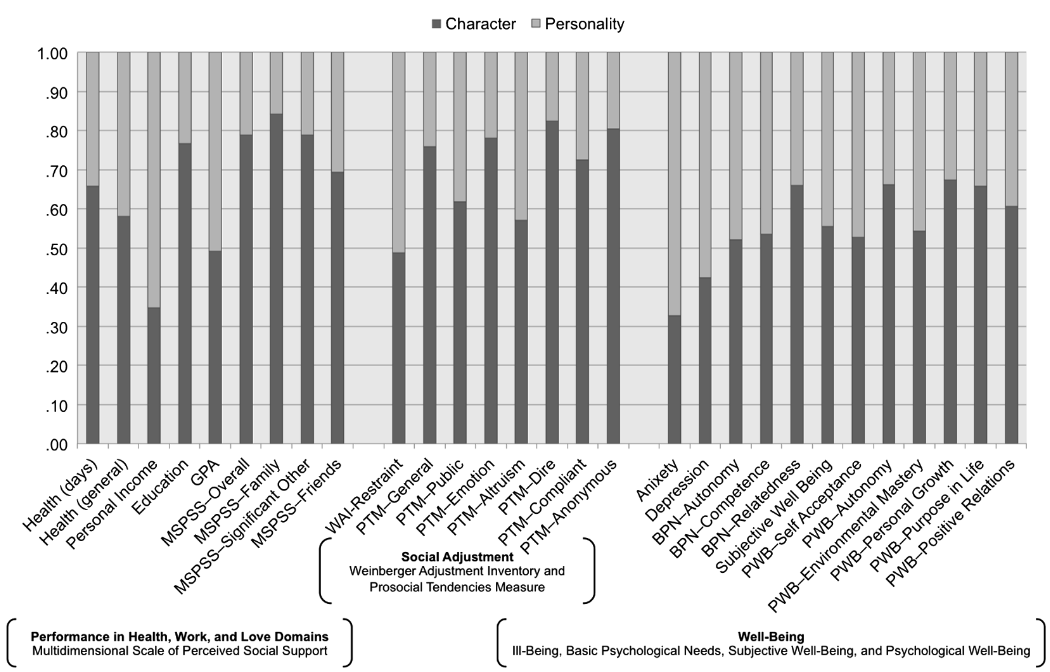

Figure 4. Proportion of explained variance in criteria attributable to character vs. Big Five personality.

(CFI = 0.85, TLI = 0.90, RMSEA = 0.03; Marsh et al., 2010) and 15-item (CFI = 0.96, TLI = 0.89, RMSEA = 0.05; Marsh, Nagengast, & Morin, 2013) measures.

In terms of substantive findings, the remaining studies collectively demonstrated support for the integrative conceptualization of character we have proposed. Study 3 established that character traits are indeed moderately to highly associated with personality at the facet-level, but were nevertheless still yet more highly related on the constructs tapping the same or similar evaluative construct. In Study 4 we found that relative to personality, character as a set of character cores generally explained more variance in moral constructs (i.e. moral identity, moral foundations, and values), whereas personality explained more variance in the more morally-neutral constructs of positive and negative affectivity. This further locates character's position in its nomological network as being more relevant to moral-evaluative constructs than Big Five personality. The results of Study 5 underscored this distinction by showing how character and Big Five personality differentially explained the relative proportion of predicted variance in outcomes as a function of how *eudaimonic* or expressive of flourishing they were. Character cores collectively predicted relevant objective outcomes across life domains, but they explained more variance in outcomes constituting flourishing such as social support and physical health than personality, which in contrast explained more variance in performance outcomes (i.e. GPA, income). Personality explained more variance in social adjustment or conforming to societal norms, whereas character explained more variance in motivations underlying prosocial tendencies. Further still, while personality explained relatively more variance in mental ill-being (i.e. depression and anxiety), character explained as much or relatively more variance in a range of well-being (i.e. basic psychological needs, subjective well-being, and PWB) constructs that collectively could be argued to constitute flourishing. In sum, these results lend credence to the idea that the CIVIC captures *virtuous* manifestations of character that engender personal and societal thriving.

Theoretical implications

Because personality taxonomies were developed to summarize variability in thoughts, feelings, and actions, it is unlikely they represent a completely full account of personality traits that encompasses all ways of being (Petrides, Pita, & Kokkinaki, 2007). It is clear that character cores and personality domains share some of the same latent structure space, but are still more or less distinguishable depending on the level of analysis and which dimensions are being examined. Paunonen and Jackson's (2000) re-analysis of Saucier and Goldberg's (1998) data showed that many adjective clusters did not fit well within the Big Five and could be coherently described using other dimensions (e.g. religious, devout, reverent; honest, ethical, moral; humorous, witty, amusing), demonstrating that there may be other individual difference dimensions beyond the Big Five within the broader, non-cognitive-ability individual difference space. As Paunonen and Jackson (2000) stated:

> If one can identify theoretically meaningful, internally consistent classes of behavior that are able to predict socially and personally significant life criteria, then such personality dimensions are important … Moreover, if such dimensions are able to account for criterion variance not accounted for by the Big Five personality factors, then those dimensions need to be considered separately in any comprehensive description of the determinants of human behavior. (p. 833)

We believe we have provided sufficient evidence for character predicting and explaining incremental variance in a number of criteria across life domains above and beyond what personality can. As such, character and the character cores that describe them need to be considered separate from and in addition to Big Five personality, especially when trying to predict outcomes that are similarly morally evaluative.

Limitations and future directions

There were a few key limitations to the present studies. First and foremost, the time interval between Time 1 and Time 2 was only approximately seven weeks on average. While this is acceptable for examination of test–retest reliability, a longer interval would have been desirable to provide stronger evidence of predictive validity. This is especially true of the MTurk sample because of the length of time generally required to see changes in outcomes that were measured, such as highest level of education attained and income. Future studies would not only include longer time intervals on the magnitude of years, but also assess measures at multiple time points at both the macro and micro level. At the macro level, a true longitudinal study would allow for analysis of character cores' development over time and lend greater power in making inferences about their predictive validity (Peterson & Park, 2009). On the micro level, experience-sampling methods would allow for multiple assessments of behavioral manifestations of character. This would address construct questions regarding key characteristics of character (e.g. consistency of action across time) while largely side-stepping issues inherent to one-time self-reports of desirable behavior, such as impression management or self-deception (Fowers, 2014).

Another limitation is this study has to do with cultural invariance. While samples did include a broad swath of society rather than relying solely on university undergraduates, the measure was not tested on non-English speakers or individuals residing outside the US. A culturally-inclusive review of virtue is required to define its boundaries in ways that guard against ethnocentrism when applying psychology to virtue ethics (Sandage & Hill, 2001) and address concerns about arbitrariness and cultural relativism when selecting what is to be included (i.e. the 'bag of virtues' argument; Hamm, 1977; Kohlberg & Mayer, 1972). And indeed one primary reason why we based our measure on Peterson and Seligman's (2004) CSV was due to their intensive, cross-cultural review of what constitutes virtue and good character (Dahlsgaard et al., 2005). However, we did not test our measure with samples from other cultures. Future studies should consider translations of the measure into other languages and assess the factor structure that emerges when the measure is used in other cultures. Measurement equivalence should also be examined to see the degree that factor structures between cultures are comparable. This has important theoretical implications given the roots this measure has in the CSV and the assertion that the broad virtues in that taxonomy are indeed cross-culturally ubiquitous (Dahlsgaard et al., 2005).

Finally, we recognize that by using a theoretically-driven classification scheme of virtues and character traits that that we may be excluding virtues that could otherwise emerge using a bottom-up approach. For example, a replication of Cawley and colleagues' (2000) psycholexical study of virtuous personality using a modified heuristic (i.e. 'Ideally is/shows …') might uncover meaningful virtues beyond what has been identified by Dahlsgaard et al. (2005) because of the greater number and variety of indicators that would likely be included. Future studies would ideally undertake such an endeavor. It is important to note that we have captured in our measure the character traits that have been valued as being virtuous, which is a description of what has been prescribed as ideal characteristics in philosophical and religious texts (Peterson & Seligman, 2004), not necessarily what should be valued. If it is the case that there are other virtues that are not adequately represented via distinct character traits in the CIVIC, then the measure will need to be modified to accurately assess character; the inventory is comprehensive in our estimation, but could still yet prove to be incomplete.

Conclusion

The present studies have provided considerable empirical support for character cores being broad traits that are relatively stable aspects of positive individual differences. The results further indicate that these traits are associated with a myriad of extrinsically and intrinsically rewarding outcomes, both objective and subjective, beyond what can be accounted for by popular conceptions of normal personality. A comprehensive taxonomy of virtuous character complements existing personality models, giving a fuller account of what a person is by elucidating what a person can aspire to be.

Note

1. There were a few cases where some factors yielded from the EFAs were not retained. Both the character traits of honesty and judgment had arguably better fit indices with a three-factor structure. With honesty, the third factor included only items that specifically reflected behavioral consistency with one's values. These items were dropped given the literature on the lack of conceptual discrimination made between virtues and values in the management and social sciences and the indiscriminant usage of the terms (cf. Wright & Goodstein, 2007); the potential to be tapping into behavioral integrity, which can be considered an adjunctive instead of a substantive virtue (i.e. one that facilitates the enactment of substantive virtues, but are amoral themselves; Audi & Murphy, 2006; Simons, 2002); and the relative narrowness of the construct. As a result, only the two primary factors under honesty were retained. A similar decision was made with the character trait of judgment. The third factor that emerged here was comprised of only two items that both spoke to the ability to think critically. Again, given the potential to be tapping into an adjunctive virtue and the narrowness of the factor the decision was made drop these items and to only retain two factors. One final exception was with zest, which seemed to tap into a construct measuring daily enthusiasm or vigor and a general energy construct. Since the former was very close to the original conception of zest and the latter was deemed too general, morally neutral, and likely related to personality dimensions, only the set of items related to the former construct was kept and thus zest was not included in the table of character traits that resulted in multiple factors.

Disclosure statement

No potential conflict of interest was reported by the authors.

Funding

This work was supported by the John Templeton Foundation [grant number 48287].

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Appendix 1. Scales used to test for convergent validity

	Scale	Authors	Response scale	M	SD	α
Gratitude	The Gratitude Questionnaire	McCullough, Emmons, and Tsang (2002)	1–7	5.73	1.05	0.89
Love	–	–	–	–	–	–
Curiosity	The Curiosity and Exploration Inventory-II	Kashdan et al. (2009)	1–5	3.09	0.82	0.90
Love of learning	Need for Cognition Scale Short Form	Cacioppo, Petty, and Kao (1984)	−4 to 4	1.18	1.61	0.96
Creativity	Creative Personality Scale	Gough (1979)	Yes/no checklist	6.42	3.84	0.74
Appreciation of beauty	The Awe-Proneness scale of the Dispositional Positive Emotions Scales	Shiota, Keltner, and John (2006).	1–7	5.25	0.96	0.82
Zest	Subjective Vitality Scale	Ryan and Frederick (1997)	1–7	4.81	1.37	0.94
Hope	The Hope Scale	Snyder et al. (1991)	1–4	3.07	0.45	0.88
Persistence	Perseverance of Effort Subscale of the Grit Scale	Duckworth, Peterson, Matthews, and Kelly (2007).	1–5	4.06	0.67	0.88
Leadership	Directiveness Subscale of the Assertiveness Scale	Lorr and More (1980)	1–5	2.97	0.90	0.92
Social perceptiveness	–	–	–	–	–	–
Perspective	Practical Wisdom Scale	Krause and Hayward (2014)	1–4	3.30	0.40	0.78
Teamwork	Group Loyalty subscale of the Individual and Group Loyalty Scale	Beer and Watson (2009).	1–5	3.53	0.80	0.90
Humor	The Trait Cheerfulness subscale of the State-Trait-Cheerfulness-Inventory	Ruch, Köhler, and van Thriel (1996))	1–4	3.10	0.53	0.92
Trustworthiness	–	–	–	–	–	–
Authenticity	Authentic Living subscale of the Authentic Living Scale	Wood, Linley, Maltby, Baliousis, and Joseph (2008)	1–7	6.14	0.79	0.87
Bravery	Social/Moral subscale of the Woodard Pury Courage Scale	Woodard and Pury (2007)	1–5	3.87	0.55	0.50
Humility	Modesty subscale of the HEXACO Personality Inventory	Lee and Ashton (2004)	1–5	4.00	0.78	0.79
Carefulness	–	–	–	–	–	–
Forethought	Consideration of Future Consequences Scale	Strathman, Gleicher, Boninger, and Edwards (1994)	1–5	3.64	0.65	0.90
Self-control	Deliberate/Nonimpulsive subscale of the Self-Control Scale	Tangney et al. (2004)	1–5	3.79	0.66	0.84
Emotional awareness	Clarity subscale of the Trait Meta-Mood Scale	Salovey, Mayer, Goldman, Turvey, and Palfai (1995)				
Spirituality	Prayer Fulfillment Subscale of the Spiritual Transcendence Scale	Piedmont (1999)	1–5	2.75	1.05	0.93
Meaning/purpose	Universality Subscale of the Spiritual Transcendence Scale	Piedmont (1999)	1–5	3.50	1.00	0.94
Perspective-taking	Perspective-Taking subscale of the Interpersonal Reactivity Index	Davis (1983)	1–5	2.90	0.67	0.84
Forgiveness	'Other' Subscale of the Heartland Forgiveness Scale	Thompson et al. (2005)	1–7	4.18	0.96	0.88
Fairness	–	–	–	–	–	–
Openness to evidence	–	–	–	–	–	–
Kindness	Santa Clara Compassion Scale	Hwang, Plante, and Lackey (2008)	1–7	5.27	1.19	0.92

Appendix 2. Descriptives and correlations of character traits

Character traits	M	SD	α	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
1. Gratitude	3.31	0.49	0.90																												
2. Love	3.47	0.49	0.83	0.62																											
3. Curiosity	3.31	0.50	0.85	0.39	0.34																										
4. Love of learning	3.34	0.50	0.94	0.42	0.36	0.86																									
5. Creativity	3.03	0.64	0.88	0.35	0.27	0.54	0.53																								
6. Appreciation of beauty	3.29	0.52	0.93	0.58	0.43	0.54	0.52	0.44																							
7. Zest	2.91	0.57	0.95	0.60	0.41	0.36	0.43	0.38	0.40																						
8. Hope	3.09	0.61	0.94	0.61	0.42	0.33	0.38	0.35	0.35	0.79																					
9. Persistence	3.14	0.50	0.96	0.56	0.41	0.37	0.49	0.38	0.37	0.70	0.61																				
10. Teamwork	2.82	0.58	0.92	0.37	0.28	0.30	0.35	0.37	0.27	0.57	0.47	0.53																			
11. Leadership	3.09	0.45	0.89	0.50	0.39	0.41	0.46	0.47	0.41	0.52	0.48	0.58	0.61																		
12. Perspective	3.07	0.53	0.89	0.38	0.35	0.32	0.33	0.34	0.32	0.34	0.31	0.42	0.42	0.58																	
13. Social perceptiveness	3.12	0.46	0.88	0.50	0.45	0.33	0.40	0.30	0.36	0.54	0.46	0.61	0.64	0.56	0.44																
14. Humor	2.98	0.65	0.89	0.29	0.25	0.27	0.26	0.34	0.21	0.31	0.27	0.25	0.37	0.41	0.34	0.30															
15. Trustworthiness	3.40	0.45	0.86	0.53	0.53	0.31	0.40	0.25	0.37	0.47	0.41	0.68	0.37	0.50	0.40	0.56	0.24														
16. Authenticity	3.34	0.48	0.85	0.52	0.47	0.34	0.40	0.26	0.40	0.40	0.37	0.54	0.27	0.45	0.33	0.41	0.22	0.64													
17. Bravery	3.21	0.49	0.86	0.53	0.41	0.43	0.50	0.35	0.47	0.47	0.42	0.62	0.43	0.53	0.43	0.47	0.28	0.59	0.62												
18. Humility	3.26	0.53	0.91	0.25	0.26	0.11	0.14	0.06	0.20	0.06	0.03	0.20	−0.02	0.13	0.21	0.18	−0.03	0.35	0.41	0.25											
19. Carefulness	3.29	0.49	0.82	0.37	0.35	0.32	0.36	0.27	0.29	0.27	0.26	0.49	0.26	0.42	0.41	0.38	0.21	0.48	0.41	0.36	0.29										
20. Forethought	3.19	0.48	0.80	0.51	0.41	0.42	0.48	0.37	0.38	0.49	0.46	0.64	0.40	0.52	0.41	0.52	0.25	0.52	0.46	0.48	0.22	0.70									
21. Self-control	3.12	0.49	0.84	0.36	0.25	0.29	0.33	0.24	0.25	0.41	0.34	0.59	0.32	0.40	0.31	0.38	0.17	0.47	0.42	0.41	0.27	0.51	0.57								
22. Emotional awareness	3.21	0.48	0.85	0.52	0.46	0.39	0.39	0.35	0.44	0.43	0.41	0.50	0.38	0.56	0.55	0.46	0.29	0.49	0.50	0.47	0.20	0.45	0.49	0.42							
23. Spirituality	2.59	0.88	0.95	0.40	0.15	0.10	0.14	0.17	0.28	0.32	0.34	0.22	0.14	0.26	0.17	0.20	0.10	0.15	0.18	0.22	0.09	0.05	0.18	0.14	0.17						
24.Meaning/purpose	2.79	0.82	0.96	0.50	0.26	0.11	0.16	0.19	0.32	0.44	0.46	0.35	0.26	0.33	0.24	0.33	0.14	0.26	0.25	0.30	0.13	0.11	0.25	0.18	0.23	0.82					
25. Perspective-taking	3.17	0.45	0.87	0.44	0.41	0.43	0.48	0.33	0.41	0.37	0.32	0.43	0.30	0.38	0.35	0.46	0.17	0.42	0.47	0.42	0.36	0.39	0.42	0.33	0.42	0.14	0.19				
26. Forgiveness	3.09	0.51	0.91	0.43	0.36	0.24	0.27	0.21	0.32	0.33	0.32	0.31	0.19	0.26	0.19	0.37	0.14	0.29	0.35	0.31	0.28	0.25	0.29	0.26	0.32	0.28	0.28	0.53			
27. Fairness	3.37	0.41	0.88	0.51	0.49	0.46	0.53	0.33	0.49	0.33	0.31	0.49	0.27	0.42	0.36	0.50	0.22	0.58	0.59	0.54	0.40	0.50	0.50	0.39	0.52	0.09	0.16	0.67	0.49		
28.Openness to evidence	3.15	0.45	0.84	0.31	0.32	0.45	0.48	0.30	0.34	0.27	0.23	0.33	0.22	0.32	0.32	0.33	0.16	0.34	0.37	0.37	0.24	0.33	0.37	0.33	0.41	−0.02	0.01	0.65	0.37	0.57	
29. Kindness	3.13	0.54	0.92	0.53	0.46	0.36	0.41	0.30	0.48	0.41	0.33	0.43	0.41	0.49	0.42	0.56	0.27	0.46	0.41	0.52	0.26	0.27	0.42	0.28	0.41	0.32	0.38	0.49	0.45	0.54	0.35

Notes: correlations ± 0.08 are significant at $p < 0.01$ and correlations ± 0.06 significant at $p < 0.05$. M = mean, SD = standard deviation, α = Cronbach's alpha.